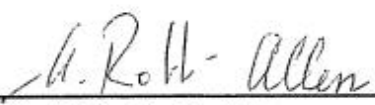
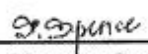


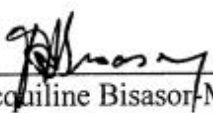
GUIDELINES FOR THE MANAGEMENT OF HIV IN THE PAEDIATRIC AND ADOLESCENT POPULATION

Ministry of Health & Wellness
Jamaica

DECEMBER 2021


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Illustrations and some content are from ICAP (International Center for AIDS Care and Treatment Programmes) Columbia University, Mailman School of Public Health with permission.

GLOSSARY OF TERMS

Glossary of Terms

3TC	Lamivudine
ABC	Abacavir
AIDS	Acquired Immune Deficiency Syndrome
ALHIV	Adolescents Living with HIV
ALT	Alanine Aminotransferase
ANC	Antenatal Clinic
APGAR	Appearance, Pulse, Grimace, Activity, Respiration
ARV	Antiretroviral Therapy
ARV	Antiretroviral
ASRH	Adolescent Sexual and Reproductive Health Services
AST	Aspartate Aminotransferase
ATV/r	Atazanavir/Ritonavir
AZT	Zidovudine
CBC	Complete Blood Count
CD4	T-Lymphocyte CD4 cell count
CMV	Cytomegalovirus
CNS	Central Nervous System
COBI	Cobicistat
CTX	Cefotaxime
DOT	Directly Observed Therapy
DRESS	Drug reaction (or rash) with Eosinophilia and Systemic Symptoms
DRV	Darunavir
DTG	Dolutegravir
EBV	Epstein-Barr Virus
EFV	Efavirenz
eGFR	Estimated Glomerular Filtration Rate

ETV	Etravirine
FPG	Fasting Plasma Glucose
FTC	Emtricitabine
GFR	Glomerular Filtration Rate
HAV	Hepatitis A Virus
HBsAg	Hepatitis B Surface Antigen
HBV	Hepatitis B Virus
HCV	Hepatitis C Virus
HCW	Health Care Workers
HIV	Human Immunodeficiency Virus
HIVAN	Human Immunodeficiency Virus Associated Nephropathy
HIVDR	HIV Drug Resistance
HPV	Human Papillomavirus
HSR	Hypersensitivity Reaction
INH	Isoniazid
INSTI	Integrase Strand Transfer Inhibitor
IPT	Isoniazid Preventive Therapy
IRIS	Immune Reconstitution Inflammatory Syndrome
IUD	Intra-Uterine Device
LBW	Low Birth Weight
LFT	Liver Function Test
LPV/r	Lopinavir/Ritonavir
MSM	Men Who Have Sex with Men
MTCT	Mother-to-Child Transmission (of HIV)
MVC	Maraviroc
NGO	Non-Governmental Organization
NNRTI	Non-Nucleoside Reverse Transcriptase Inhibitor
NRTI	Nucleoside Reverse Transcriptase Inhibitor
NVP	Nevirapine

OI	Opportunistic Infection
PEP	Post-Exposure Prophylaxis
PHIVA	Perinatally Human Immuno-Deficiency Virus-Infected Adolescents
PI	Protease Inhibitor
PCP	<i>Pneumocystis jirovecii</i> (Formally <i>carinii</i>) Pneumonia
PML	Progressive Multifocal Leukoencephalopathy
PMTCT	Prevention of Mother-to-Child Transmission (of HIV)
PO	Per os
POPD	Paediatric Outpatient Department
PROM	Premature Rupture of Membranes
RAL	Raltegravir
RBG	Random Blood Glucose
RNA	Ribonucleic Acid
RPV	Rilpivirine
RTV	Ritonavir
sdNVP	Single-Dose Nevirapine
SJS/TEN/EM	Stevens-Johnson Syndrome/Toxic Epidermal Necrolysis/Erythema
SRH	Sexual and Reproductive Health
STI	Sexually Transmitted Infection
TAF	Tenofovir Alafenamide
TB	Tuberculosis
TC	Total Cholesterol
TDF	Tenofovir Disoproxil Fumarate
TG	Triglycerides
TMP-SMX	Trimethoprim-Sulphamethoxazole
ULN	Upper Limits of Normal
UNICEF	United Nations Children's Fund
WHO	World Health Organization

INTRODUCTION

Introduction

Rationale

Jamaica has made great strides in the care of infants, children, and adolescents affected by HIV /AIDS over the past 20 years. Successful interventions resulted in reduced vertical transmission of HIV, approaching virtual elimination targets and increased survival of perinatally affected children.

Challenges peculiar to children and adolescents affected by HIV impact their viral suppression and require targeted strategies tailored for this vulnerable population. Local research highlights early transition to adult care, psychosocial issues, and challenges affecting caregivers of HIV infected children, impact medication adherence. On a global scale, adolescents and young adults, are less likely than older counterparts to be retained in care and on ARV and have worse clinical outcomes. Adolescent-centered services and family-focused interventions are needed to ensure high virologic suppression rates. International partners have set 95-95-95 global targets aimed at 95% tested, 95% linked and 95% virologic suppressed towards ending the growing incidence, economic and health burden of this epidemic. In keeping pace with updated treatment guidelines and increased accessibility to new therapeutic agents, this manual serves to integrate international standards within a Jamaican context, for the standardized care and wellbeing of infants, children, adolescents, and families affected by HIV. New ARV regimens are complemented with supportive strategies such as pill swallowing, disclosure, adherence, and psychosocial support, adolescent and family-focused care.

Guideline Development

The National treatment and care guidelines for HIV infected children and adolescents was co-developed with key stakeholders who suggested a framework for the content in 2019. Information for the content was compiled using several resources. These include guidelines from the World Health Organization, National Institute of Health, Centres for Disease Control, training manuals, international standardized care guidelines for children and adolescents, opinions of local and international experts along with locally published works. Drafts of the manual were reviewed with HIV experts, paediatricians and other allied health specialists working in the field of HIV and revised to incorporate local and international best practices.

Audience

The target audiences are all clinicians, nurses and allied specialists working in the field of HIV. The content is geared at an overall appreciation of the standards required in all areas of team management.

GUIDELINE: INTRODUCTION		
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Approved by: Senior Medical Officer, HSTU		

CHAPTER 1: HIV EXPOSED INFANT

1.1 Protocol for the HIV Infected Mother

1.2 Definition of a High Risk Mother

1.3 Guideline for the Newborn

1.4 Chemoprophylaxis

1.1 Protocol for the HIV Infected Mother

- ARV is to be commenced as soon as possible if HIV infection is diagnosed during pregnancy and continued for life (Recommended ARV: Tenofovir Lamivudine Dolutegravir/Efavirenz; see PMTCT manual)
- Continue and optimize ARV for those known HIV infected women already commenced on ARV; reinforce adherence to promote maximal viral suppression
- Scheduled caesarean delivery is recommended for prevention of perinatal transmission in women who have received antepartum ARV drugs but who have **detectable viremia (HIV RNA >1000 copies/mL)** near the time of delivery
- In addition to ART's, consider emergency Caesarean Section for mothers who are diagnosed very late in pregnancy, who have extremely high-risk factors and those without a known viral load
- Advise the mother to apply cold compress on the breast and not to express breast milk; instead, she should leave the breast unstimulated and well supported

1.2 Definition of a High-Risk and Low-Risk Mother

The High-Risk Mother

- A woman who is poorly adherent or non-adherent to ARV before, and during pregnancy
- New HIV diagnosis in a pregnant or breastfeeding woman with a prior negative HIV test during pregnancy
- HIV exposure (maternal HIV infection) first identified at delivery or in the postpartum period in a breastfed infant; if the rapid test is positive, indeterminate or inconclusive or HIV status is unknown, then manage as high risk
- If viral load testing is not available, pregnant HIV infected woman commencing ARV (or switching ARV due to virologic failure) within 4 weeks before delivery
- If viral load testing is available, pregnant women whose viral load exceeds 1000 copies/ mL within 4 weeks before delivery

Post Delivery Care of the High Risk Mother and Infant

- Ensure universal precautions
- Ensure infant has limited contact time with maternal blood especially if viral suppression is suboptimal, newly diagnosed at delivery or status unknown
- Ensure decontamination of newborn to limit exposure to potentially infectious birth products and secretions
- Breastfeeding is not recommended in the HIV positive mother

The Low-Risk Mother

- Mother who received > 4 weeks of ARVs at the time of delivery, who has an established HIV infection, who is virally suppressed (viral load <1000 copies) and who has not breastfed

1.3 Guidelines for the Newborn

- At birth thorough maternal history and evaluation of the newborn is done by a member of the paediatric team
- HIV-infected mothers may be co-infected with other pathogens that may be transmitted from mother to child, such as cytomegalovirus, herpes simplex virus, hepatitis B, hepatitis C, syphilis, toxoplasmosis, or tuberculosis. Healthcare providers must carefully consider these in the evaluation of the infant.
- Immediately after delivery (within 6 hours of birth) commence chemoprophylaxis according to protocol and give a prescription for 6 weeks which must be filled before discharge.
- Advise mother on breast milk replacement and refer the dietary department for infant formula collection and counselling, before discharge
- Clinicians caring for a woman with HIV who is considering breastfeeding against strong recommendation, should consult with an expert (see **Appendix 1** re special circumstances for breastfeeding).
- Give a 2-week appointment to POPD to ensure compliance with infant prophylaxis and identify barriers to care.

1.4 Chemoprophylaxis

For Infant of a Low-Risk Mother*

- Zidovudine 4 mg/kg /dose every 12 hours for 6 weeks. See chart for premature newborn

AND

- Nevirapine stat 2 mg/kg/dose

For Infant of a High-Risk Mother

Dual prophylaxis with:

- Zidovudine 4 mg/kg /dose every 12 hours for 6 weeks. See chart for premature newborn

AND

- Nevirapine 2 mg/kg/dose once daily for 6 weeks

Table 1: Zidovudine Dose and Schedule of Administration by Gestational Age

GESTATIONAL AGE	MEDICATIONS
≥35 weeks gestation at birth	Zidovudine 4 mg/kg/dose PO twice daily for 6 weeks
30 to < 35 weeks gestation at birth	Zidovudine 2 mg/kg/dose every 12 hours for 2 weeks Then Zidovudine 3mg/kg/dose every 12 hours for 4 weeks
< 30 weeks gestation at birth	Zidovudine 2 mg/kg/dose every 12 hours for 4 weeks Then Zidovudine 3mg/kg/dose every 12 hours for 2 weeks

***See PMTCT Manual for further details**

GUIDELINE: HIV EXPOSED INFANT		
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CHAPTER 2: HIV DIAGNOSIS IN INFANTS AND ADOLESCENTS

2.1 History and Laboratory Confirmation

2.2 Historical Highlights for the HIV Exposed Child

2.3 The Suspected HIV Infected Child

2.4 HIV Diagnosis

2.1 History and Laboratory Confirmation

HIV infection should be suspected in infants or children in the following circumstances:

- Unknown maternal HIV status at delivery
- Maternal history of HIV infection
- A history of sexual abuse or exposure to contaminated needles; or
- Presentation with symptoms or medical conditions that could indicate HIV infection, including malnutrition or presumed and confirmed TB cases.

Appendix 2 outlines the CDC clinical and immune categories of HIV infection in children.

Features are classified as:

- Mildly symptomatic (A)
- Moderately symptomatic (B)
- AIDS-defining (C)

Table 2: Common clinical manifestations in Jamaican Infants and Children by CDC disease severity

CATEGORY	CLINICAL MANIFESTATIONS
Category A	lymphadenopathy, hepatosplenomegaly, dermatitis, and persistent or recurrent upper respiratory tract infections
Category B	bacterial sepsis, oropharyngeal candidiasis, and recurrent diarrhoea
Category C	serious infections delayed milestones, regression of milestones, failure to thrive

2.2 Historical Highlights for the HIV Exposed Child

Review perinatal history to determine the risk of mother-to-child-transmission of HIV infection, including:

- Timing of maternal ARV initiation
- Adherence to ARV during pregnancy
- Antenatal complications, e.g., infections
- Labour and delivery: gestational age, duration, premature rupture of membranes (PROM), instrumentation, mode of delivery, intrapartum ARV
- At birth: birth weight, height, head circumference, APGAR score/perinatal asphyxia; neonatal complications
- Infant ARV chemoprophylaxis: Initiation and compliance with Zidovudine and Nevirapine chemoprophylaxis in the newborn
- Infant feeding: breast milk replacement and/or breast milk exposure
- Pertinent open-ended questions should be used, such as:
 1. *“How many times did you miss your medication during pregnancy?”*
 2. *“How many times did you breastfeed?”*
- Enquire how much of the Zidovudine was measured. If mom is uncertain, this should raise a red flag
- Disclosure issues: determine if other siblings will need to be tested; inquire if the mother’s partner was tested and refer for testing and counselling accordingly.
- Review immunization records and do a detailed assessment of growth and development, nutrition, and history of concurrent illnesses (e.g., bacterial sepsis, hospitalizations, severe dermatitis, thrush, diarrhoea, weight loss, poor weight gain).
- Review ARV chemoprophylaxis scheduling and dosing
- Ensure PCP chemoprophylaxis is commenced at 6 weeks of age with trimethoprim-sulphamethoxazole (Bactrim®) [at trimethoprim 5 mg/kg per dose daily or dosed Mon, Wed, Fri on recommendation by the consultant paediatrician]
- Refer mother to Treatment Care and Support Officer

2.3 The Suspected HIV Infected Child

History should include:

- Perinatal history and risk for mother-to-child-transmission
- Mode of transmission: MTCT, behavioural (sexual – forced or consensual, intravenous drug use), needle stick injury
- History of illnesses and hospitalizations (e.g., bacterial sepsis, TB, recurrent diarrhoea, recurrent ear infections and sinusitis, severe dermatitis, thrush, failure to thrive, delayed milestones and/or regression; see CDC clinical features in Table 2); past medical history
- If exposed to ARVs: ARV history, regimens, adherence, interruptions, adverse effects
- History of linkage to and retention in care at treatment sites
- Disclosure
- Family history of HIV infection, social history and barriers to care

The history should be followed by a thorough examination.

2.4 HIV Diagnosis

• Infants born to low-risk mothers should have 2 HIV DNA PCR tests at six weeks and four months PLUS a rapid diagnostic test at 18 months • Infants born to high-risk mothers should have 3 HIV DNA PCR tests at 2 weeks, 6 weeks and 4 months PLUS a rapid diagnostic test at 18 months.
• Any infant with an initial positive virological test result (positive HIV DNA PCR), should be commenced on ARV without delay, irrespective of CD4 count. Pertinent baseline investigations, should be done at the time of ARV initiation and the testing algorithm followed as above.
• Infants under 18 months with signs or symptoms suggestive of HIV infection should undergo virological testing (HIV DNA PCR). HIV serological testing should be done at 18 months to confirm the diagnosis.
• Children (18 months or older) with suspected HIV infection or HIV exposure should have a rapid diagnostic test for HIV.
• Infants and children with unknown HIV status who are diagnosed with malnutrition or TB should be routinely tested for HIV.
• Repeat antibody testing for late seroreverters. These are perinatally HIV exposed, non-breastfed infants with no virologic or clinical evidence of HIV, but with an indeterminate HIV serology at 18-24 months.
• Residual maternal antibodies may continue to circulate up to 24 months. The HIV exposed infants who have repeated positive HIV tests at age 18 months to 24 months require a repeat HIV DNA PCR (or other virological tests) to definitively exclude or confirm HIV infection.
• Diagnosis of HIV-1 infection in infants and children >24 months requires an HIV antibody test.
• Definitive exclusion of HIV infection in a non-breastfed infant is based on two or more negative PCR tests; one at age ≥ 1 month and one at age ≥ 4 months and confirmed with a negative serologic test >18 months.
• For the woman who chooses to breast feed see Appendix 1. The exclusively breastfed HIV exposed infants should be maintained on Nevirapine prophylaxis and have PCR testing every 3 months throughout breastfeeding and 6 weeks, 3 months, and 6 months after breastmilk cessation PLUS a rapid test at 18 to 24 months of age.

GUIDELINE: HIV DIAGNOSIS IN INFANTS AND ADOLESCENTS		
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CHAPTER 3: MEDICAL MANAGEMENT OF HIV IN CHILDREN

3.1 Routine Care

3.2 Immunization

3.3 Nutrition and Healthy Living

3.4 Neurocognitive and Developmental Assessment

3.5 Laboratory Tests

Medical Management of HIV in Children

The diagnosis of HIV infection in a child is very devastating for a family. Clinicians should aim to establish a long-term relationship with the patient and the family. The family should be assessed by the medical social worker for addressing their issues around social service or the need for financial support. Members of the family should be offered referral to the clinical psychologist. The establishment of trust and rapport greatly improves adherence to medical treatment.

3.1 Routine Care

At each visit:

- The history for the PLHIV should be executed as outlined in Chapter 2
- Immunization records, growth and development should be evaluated at all ages and stages through to adolescence
- A complete physical examination should be done to assess for signs and symptoms associated with HIV disease
- Antiretroviral medications should be reviewed, and doses adjusted for growth
- Adherence/Compliance assessed (see Adherence)
- Refer caregivers for support if challenges identified in administering medications
- Ensure that biological caregivers (mother, father) and siblings are tested, and linked to care; identify barriers to care
- Encourage involvement of adolescent and parents/ guardians in support groups

3.2 Immunization

The immunization schedule follows the recommended Jamaica Ministry of Health and Wellness guidelines with added recommendations.

Table 3: Immunization Schedule

Vaccine	Dose Schedule
Bacillus Calmette–Guérin (BCG)	Birth-6 weeks
Diphtheria Pertussis Tetanus (DPT) Inactivated Polio Vaccine (IPV)	6 weeks 3 months 6 months 18 months 4-6 years
Hepatitis B (HepB)	6 weeks 3 months 6 months
Haemophilus Influenza Type B (Hib)	6 weeks 3 months 6 months
Pneumococcal	6 weeks 3 months 6 months
Measles, Mumps, Rubella (MMR)	12 months 18 months
DT (or Boostrix-Tdap)	11-12 years
Human Papilloma Virus (HPV)(girls)	9-14 or (11-12 years Grade 7 cohort) Bivalent 3-dose vaccination schedule: 2 nd dose 1-2 months after 1 st ; 3 rd dose 6 months after 1 st

Vaccine	Dose Schedule
Varicella-zoster*	12 months 18 months
Rotavirus*	2 doses of Rotarix® between 2 and 6 months of age, 2 months apart
Influenza*	Annually
*Not currently part of the MOHW Immunization Schedule/Guidelines, but highly recommended for HIV infected infants Hepatitis B should be given at birth to hepatitis B surface antigen positive mothers or as per the routine schedule NB. Live vaccines (MMR, varicella, administration is not advised if severely immunocompromised (%CD4 < 15%; absolute CD4 < 200 cells/μL in those > 5 years)	

BCG

BCG should be administered to all HIV exposed newborns.

For high risk newborns (see definition in infant prophylaxis section) on dual prophylaxis, one may delay administration of BCG after 2 negative HIV DNA PCR results (done at 2 weeks and 6 weeks) to reduce the risk of disseminated vaccine-related BCG in a potential HIV infected newborn.

MMR

Two doses of MMR vaccine are recommended for all HIV-infected individuals aged ≥12 months who do not have evidence of current severe immunosuppression. Doses should be given at least six (6) months apart and is recommended to be given at 12 and 18 months.

Pneumococcal

Pneumococcal conjugate vaccines are recommended (2 doses) for HIV infected children and adolescents who were not previously vaccinated. For HIV adolescents not previously immunised with the pneumococcal vaccine, administer one dose of 13-valent pneumococcal conjugate vaccine (PCV13) followed by one dose of 23-valent pneumococcal polysaccharide vaccine (PPSV23) at least 2 months later.

Hepatitis B Vaccine

Administer a 3-dose series of single-antigen hepatitis B vaccine (HepB) 0, 1, and 6 months for previously unvaccinated adolescents.

HPV

Currently, the National Immunization Programme offers the bivalent HPV vaccine to girls aged 9-14 years.

HIV infected females 9 through 26 years should be vaccinated with HPV as cervical cancer outcomes are worse in HIV infected females. A 3-dose schedule (0, 1–2, 6 months) should be used for all vaccinations initiated ≥ 15 years of age and in those younger than 15 years known to be immunocompromised.

Immunization of HIV infected males age 9-26 years is recommended.

Live vaccine administration should be delayed in children with severe immunosuppression, and until there is improvement in immune status with use of ARV.

Severe immunosuppression is defined as ≥ 6 months below the age-specific CD4 count criteria:

- CD4 count < 750 cells/ μ L and percentage $< 15\%$ while aged ≤ 12 months;
- CD4 count < 500 cells/ μ L and percentage $< 15\%$ while aged 1 through 5 years;
- CD4 count < 200 cells/ μ L and percentage $< 15\%$ for individuals aged > 5 years

Varicella-zoster

Two doses of varicella vaccine are recommended for all HIV-infected individuals aged ≥ 12 months who do not have evidence of current severe immunosuppression, and who do not have medical/serological proof of past infection.

Managing the oral health for these HIV-infected children requires close synchronization between the dentist, the pedodontist, the paediatrician, the nutritionist, and the child's parents or caregivers.

HIV-infected children should receive a periodic oral and dental examination and hygienic care every six months (or more frequently as indicated). Oral diseases can be recognized early through examinations, monitored and treated. Early recognition and management of oral condition reduce disease severity and ultimately reduces clinical costs.

Oral prophylaxis i.e. dental cleaning and oral hygiene instruction must be provided every three to six months and the regular use of fluoride gels should be implemented based on the status of enamel/dentin health.

Moderately and severely immunosuppressed children (Appendix 2: CDC immune classification) and those with an absolute neutrophil count below 1,500/mm³ and/or with deranged liver function tests will need antibiotic prophylaxis for routine dental care. Clients with low platelet quantities (10,000 - 30,000/ μ L) require a platelet transfusion before major surgical procedures.

Aphthous ulcers occur typically in severely immunocompromised patients and can be treated with magic mouthwash (diphenhydramine, viscous lidocaine, antacid, nystatin and corticosteroids made up by the pharmacist) to swish and swallow.

Table 4: EC-Clearing House and WHO classification of oral manifestations of paediatric human immunodeficiency virus (HIV) disease.

<i>Group 1 lesions commonly associated with pediatric HIV infection</i>	<i>Group 2 lesions less commonly associated with pediatric HIV infection</i>	<i>Group 3 lesions strongly associated with HIV infection but rare in children</i>
Candidiasis	Seborrheic dermatitis	Neoplasms
Erythematous	Bacterial infections of oral tissues	Kaposi's sarcoma
Pseudomembranous	Necrotizing (ulcerative) stomatitis	Non-Hodgkin's lymphoma
Angular cheilitis	Periodontal diseases	Oral hairy leukoplakia
Herpes simplex virus infection	Necrotizing (ulcerative) gingivitis	Tuberculosis-related ulcers
Linear gingival erythema	Necrotizing (ulcerative) periodontitis	
Parotid enlargement	Viral infections	
Recurrent aphthous ulcers (Fig. 1)	Cytomegalovirus	
Minor	Human papilloma virus	
Major	Molluscum contagiosum	
Herpetiform	Varicella zoster virus	
	Herpes zoster	
	Varicella	
	Xerostomia	

3.3 Nutrition and Healthy Living

A healthy lifestyle should be encouraged in children and adolescents.

A nutritional history should be ascertained at each clinic visit and growth parameters plotted on growth charts for interpretation.

Growth delay may be the first sign of symptomatic HIV disease, and in older children, weight loss and wasting occur with progressive HIV disease.

Conversely, long term use of specific antiretrovirals may be associated with lipodystrophy (fat maldistribution) and derangements in glucose and lipids which increase the risk of cardiovascular disease. Lifestyle modification such as implementing a diet low in saturated fat, cholesterol, trans fat, and refined sugars; increasing physical activity and ceasing smoking, can mitigate and help manage these adverse events.

At minimum, a nutritionist should see all PHIVA annually.

Health workers should counsel and educate clients and their family members about good nutrition. Patients should be advised to:

- Eat a well-balanced diet that includes a variety of fresh foods, and that is based on what is locally available and affordable
 - Make “energy-giving foods” (“GO” foods/starches) the biggest part of every meal
 - Eat “body building foods” (“GROW” foods) with every meal. Whenever possible, a meal should contain some meat, fish, or other foods from animals. When properly combined, plant proteins (beans, peas, soya beans, peanuts, and other nuts) also provide good quality protein
 - Eat “protective foods” (“GLOW” foods), which include all kinds of fruits and vegetables, every day
 - Use fats and oils in moderation
 - Avoid junk food and processed foods
- Increase caloric intake, especially with “GO” and “GROW” foods
 - Asymptomatic ALHIV may need 10–15% more energy intake than people without HIV

- Symptomatic ALHIV require between 20–30% more energy intake, this translates to another full meal each day or 2–3 additional snacks
- Try to eat small meals frequently and have a regular meal schedule especially adolescents who often have erratic eating patterns due to their busy lives
- Take a daily multivitamin supplement as a routine part of care to prevent micronutrient deficiencies
- Enquire about food security issues. If there is limited food access, engage a multidisciplinary team-based approach to address social issues and link families to social support networks such as Programme of Advancement Through Health and Education (PATH), Food for the Poor, or organizations to assist with small business development or gardening to grow food.

Other healthy habits that complement good nutrition:

- Always attending appointments and taking antiretroviral medication while reporting any health-related changes at each visit
- Practice good food hygiene such as washing fruits and vegetables; thorough cooking of meat, poultry, and fish; using clean utensils and clean preparation surfaces; covering and proper storage of food
- Practicing good personal hygiene such as regular handwashing and bathing
- Staying active with regular physical exercise. That is 60 minutes of moderate to vigorous daily physical activity for children and adolescents ages 6-17 years
- Stop smoking. People who smoke increase their risk for heart disease, cancer, serious lung diseases and infections such as pneumonia, and other illnesses. People with HIV are more likely to develop these harmful consequences of smoking than those without HIV. Discourage adolescents from developing a habit of smoking
- Get adequate rest. The table below outlines the amount of rest required by age group

Table 5: Rest Requirement by Age Group

Age /Years	Sleep Duration/ 24 hour
1-2	11-14 hours including naps
3-5	10-13 hours including naps
6-12	9-12 hours
13-18	8-10 hours

- Drinking adequate water. The recommended daily water intake is outlined below. However, increased amounts are required after exercise or on a hot day.

Table 6: Recommended Water Intake

Age /Years	Glasses of Water/ Day
5-8	5 (1 litre)
9-12	7 (1.5 litre)
>13	8-10 (2 litres)

3.4 Neurocognitive and Developmental Assessment

HIV infection in children, especially those perinatally infected, can have associated developmental delays and cognitive impairments. Cognitive impairments can include language, motor and behavioural impairments. Some children living with HIV have normal development, some have a mild impairment, and others have a severe impairment. Factors that affect the degree of impairment include; the timing of HIV infection, the extent of central nervous system involvement and inflammation, and the use of ARVs.

Assessment of neurocognitive and developmental status should be routinely incorporated into the care of all children and adolescents with HIV infection.

Signs and symptoms of neurocognitive and developmental disorders are:

- Slowed psychomotor speed (taking longer than normal to understand what someone else is saying and then respond)
- Delayed expressive language skills
- Memory deficits
- Poor attention
- Developmental impairment (failure to achieve developmental milestones); developmental impairment is most common among children who experienced severe immunodeficiency during the first few years of life. However, even children and youth with less advanced HIV disease can have mild to moderate developmental impairments related to HIV infection.
- Difficulty learning social behaviours and/or self-care

Management and treatment for neurocognitive and developmental disorders:

- Provide client and family with tailored supportive counselling that meets the unique strengths, disabilities, and needs of the child/adolescent
- Encourage caregivers to follow this general principle: reward effort, not results

- Ensure that the child/adolescent is on an appropriate ARV regimen to prevent or slow further progression of neurocognitive impairment
- Refer the client for neuropsychological testing, e.g., to a clinic psychologist MOHW or Mico Child Assessment and Research in Education Centre (CARE).
- If available, link client and family to specialized care and community-based resources for children and adolescents with intellectual and developmental disabilities
- Provide the caregivers of adolescents who are severely impaired with assistance and support, including through linkages to community resources
- Anticipate that neurocognitive impairment will delay the transition to adult care in older PHIVA
- Determine functional limitations and strengths, specifically, vocational, cognitive, educational and potential for life care planning.

3.5 Laboratory Test

Laboratory tests for initiating ARV and monitoring for toxicity include the following: a complete blood count (CBC), urinalysis, serum chemistry panel including electrolytes, creatinine, glucose, and hepatic transaminases. The levels of serum lipids (cholesterol, triglycerides) should also be measured.

Biochemical markers may also be used for nutritional evaluation (please see the interpretation of laboratory tests).

CD4 enumeration assists in staging the degree of immunodeficiency. Normal ranges of CD4 T cell lymphocyte (CD4) counts vary with age (See appendix 2: CDC Immune category). Mild intercurrent illness, the receipt of vaccinations, or exercise can produce a transient decrease in CD4 cell count and percentage. Therefore, CD4 cell count and percentage are best measured when patients are clinically stable.

HIV plasma RNA (viral load) determination is desirable (but not necessary) at baseline. However, it is the best biomedical marker of ARV efficacy.

All newly diagnosed infants, children, and adolescents should be screened for co-infections: VDRL, Hepatitis B, C, HTLV 1&2, toxoplasmosis, rubella, cytomegalovirus, herpes virus, Epstein Barr virus (TORCHE); gonorrhoea, varicella-zoster, Tuberculosis screening with Mantoux testing and chlamydia PCR (sexually active adolescents).

All HIV positive patients must have an initial CD4 count followed by 3 and 6 monthly tests until the patient is virally suppressed. Discontinue CD4 monitoring if the VL is undetectable and above the PJP prophylaxis threshold of >350 cells/mm³.

CBC, and chemistries can be monitored less frequently (every 6 -12 months) in children and youth who are adherent to therapy, who have CD4 cell values that are well above the threshold for OI risk, and who have had sustained virologic suppression and stable clinical status for more than 2 to 3 years.

Viral load monitoring should be done every 2 months after initiating or changing medications. Follow up every two weeks for adherence then monitor the viral load every 2 months until undetectable. After that, visits can be every 2-3 months if there are no adherence issues.

After ARV initiation VL should be <400 copies by 2-3 months and <50 copies by 6 months. Immediate assessment is warranted if expected fall does not occur.

Table 7 provides a proposed general monitoring schedule, which should be adjusted based on the specific ART regimen and response to therapy.

Table 7: Monitoring Schedule for the Paediatric Population

Lab testing	Enrolment/ ART initiation	2 weekly on therapy	Week 6 on therapy	Every 3 months	Every 6 months	When switching ARV
Medical History and examination	✓	✓	✓	✓		✓
Adherence Evaluation		✓	✓	✓		✓
CD4*	✓			✓	✓	✓
Viral Load**	✓		✓		✓	✓
CBC Dif	✓		✓		✓	✓
Chemistry/LFT	✓		✓		✓	✓
Lipid	✓				✓	
RBG	✓				✓	
Urinalysis***	✓				✓	
Resistance Testing						✓
TORCHE & Mantoux	✓					
Hepatitis B	✓					✓
Developmental examination 0-12mo 1-3 y >3y				✓	✓	
Ophthalmology****						
Dental					✓	
Gynaecologic and pregnancy test (post-pubertal)						✓

*CD4 every 3-6 months until virologically suppressed then discontinue. Retest at failure or switch

** recommend more frequent evaluation of VL at ART initiation/switch. Every 2 months until undetectable.

*** If on TDF

**** Every 6 months if severely immunocompromised

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CHAPTER 4: PREVENTION OF OPPORTUNISTIC INFECTIONS (OI)

4.1 Immune Reconstitution Inflammatory Syndrome (IRIS)

Prevention of Opportunistic Infections (OI)

In Jamaican children, oral candidiasis, pulmonary tuberculosis, *Pneumocystis jirovecii* pneumonia and pneumococcal infections have been the most frequent opportunistic infections.

ARVs reduce the incidence of OIs and improve survival, independent of the use of OI antimicrobial prophylaxis.

ARVs can lead to improvement or resolution of certain OIs, such as progressive multifocal leukoencephalopathy (PML) or microsporidiosis, for which effective specific treatments are not available.

ARVs do not replace the need for OI prophylaxis in children with severe immune suppression.

4.1 Immune Reconstitution Inflammatory Syndrome (IRIS)

Initiation of ARVs in individuals with an acute or latent OI can lead to Immune Reconstitution Inflammatory Syndrome (IRIS). This exaggerated inflammatory reaction, can present with paradoxical worsening of symptoms of an existing active OI. New appearance of an OI may occur, where infectious pathogens were previously not recognized by the immune system and can now evoke an immune response.

Treatment of IRIS has been based on the severity of the disease as follows:

- For mild cases, observation alone with close clinical and laboratory monitoring may be enough.
- For moderate cases, non-steroidal anti-inflammatory drugs have been used to ameliorate symptoms.
- For severe cases, corticosteroids such as dexamethasone have been used. During that time, ART generally is continued.

The conditions most commonly associated with IRIS in children include mycobacterial infections, herpes zoster, HSV, and cryptococcal infection.

Reactions related to Bacille Calmette-Guérin vaccine have been one of the most common IRIS manifestations in children.

Children who present with an OI due to poor virologic response (because of poor adherence, inappropriate ARV regimens, drug resistance, or some combination of these factors) represent a failure of therapy. Treatment of the OI should be initiated, adherence assessed, barriers addressed, ARV regimen reviewed and drug resistance testing performed.

Table 8 shows common opportunistic infections, their indications, and preventative drug therapy.

Table 8: Common Opportunistic Infections and their Treatment

Infection	Indication	Therapy
<i>Mycobacterium avium Complex (MAC)</i>	<u>Age(years)</u> <u>Primary Prophylaxis if</u> <1 CD4 count <750 cells/mm³ 1 to <2 CD4 count <500 cells/mm³; 2 to <6 CD4 count <75cells/mm³; ≥6years CD4 count <50 cells/mm³ <u>Age(years)</u> <u>Discontinue Prophylaxis</u> After ≥6 months of ART, <i>and</i>: 2 to <6 CD4 count >200 cells/mm³ ≥6 CD4 count >100 cells/mm³ For > 3 consecutive months Do not discontinue in HIV infected children <2 years	Clarithromycin 7.5 mg/kg body weight (maximum 500 mg) orally twice daily, OR Azithromycin 20 mg/kg body weight (maximum 1200 mg) orally once weekly
<i>Pneumocystis Jirovecii</i>	HIV indeterminate infants from aged 4–6 weeks to 12 months.	5 mg/kg body weight/dose with

Infection	Indication	Therapy
Pneumonia(PJP)	HIV-infected children <u>Age (years)</u> <u>Primary Prophylaxis if</u> 1 to <6 CD4 count <500 cells/mm³ 6–12 CD4 count <200 cells/mm³ or CD4 % <15% <u>Age(years)</u> <u>Discontinue Prophylaxis</u> - After ≥6 months of ART, <i>and</i>: 1 to <6 CD4 count ≥ 500 cells/mm³ ≥6 CD4 count ≥ 200 cells/mm³ or CD4 % ≥15% For > 3 consecutive months Do not discontinue in HIV infected children < 1years	Cotrimoxazole dosing based on trimethoprim component. • The total daily dose should not exceed 320 mg TMP and 1600 mg SMX. Can be given 3 days per week on consecutive days or on alternate days OR every day (total daily dose of TMP 5–10 mg/kg body weight given as a single dose each day) Give steroids to a child with probable PJP and O2 saturation < ~93% at rest or with exertion

Infection	Indication	Therapy
Tuberculosis	Children living with HIV who are close contacts of an active TB case and with no evidence of TB disease should begin Isoniazid Prevention Therapy regardless of age.	INH 10 mg/kg (range 7-15 mg/kg, maximum 300 mg/day) daily for at least 6 months
Fungal prophylaxis	Consider in infant/toddler with a history of candida oesophagitis For chronic thrush topical prophylaxis is preferred over systemic Use secondary prophylaxis after histoplasmosis, coccidiomycosis or cryptococcosis	Nystatin Fluconazole
Varicella (VZV)	Varicella exposure	Oral acyclovir 40-80 mg/kg/day PO for 7 days VZV Intravenous immunoglobulin (IVIG) within 4 days of VZV exposure is effective but expensive
Toxoplasmosis	Severely immunosuppressed child with toxoplasmosis positive serology may be at risk	Daily Bactrim 5 mg/kg/dose of TMP component

GUIDELINE: PREVENTION OF OPPORTUNISTIC INFECTIONS (OI)		
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CHAPTER 5: PILL SWALLOWING GUIDELINES

- 5.1 Overcoming Barriers for Swallowing, Pills and Capsules**
- 5.2 Teaching Pill Swallowing**
- 5.3 Pill Swallowing Techniques**

Pill Swallowing Guidelines

Children from about three years of age can be pill trained. The best candidates are children who have not had unpleasant experiences with pills or who have no behavioural problems with taking medication.

It is important that the trainer not give any signals that pill-taking may be difficult or unpleasant.

5.1 Overcoming Barriers for Swallowing Pills and Capsules

Anxiety – The process of swallowing is more difficult, if children are afraid of swallowing pills, as they may become tense. This tension, particularly in the throat, neck and chest can make the child feel like they are having trouble breathing which may, in itself, cause further anxiety.

Practicing pill swallowing techniques can help to reduce the child's anxiety and make them feel more relaxed, therefore increasing their chance of success. Parents may lead by example and demonstrate that swallowing pills or capsules is easy.

Strong gag reflex – Children who are fussy eaters or who frequently gag on food and drink can often struggle with swallowing medicines. Getting the child to take a deep breath before inserting a tablet or capsule in their mouth can help them to suppress their gag reflex.

Texture, size and shape – The size and shape of the tablet or capsule and the nature of the coating, can affect the ease of swallowing. Using yogurt or a thick drink, such as a milkshake, may help to reduce a child's awareness that a tablet or capsule is being swallowed.

5.2 Teaching Pill Swallowing

1. Ask the child to take a few sips of water or their favourite drink to moisten their mouth and throat
2. Start with the smallest sized candy (e.g., tic tac or cake sprinkles) and explain to the child that you are going to teach them a simple way to swallow pills
3. Show them by putting a single candy towards the back of your tongue, taking a gulp of water and swallowing the candy.
4. Tell the child it is their turn. They should keep trying up to three times if they are unable to swallow, and they can chew the candy if the third time is unsuccessful.
5. Repeat this until the child is comfortable with a particular sized candy– this usually takes about three successful tries. Then move up to a slightly larger size candy and repeat the procedure until there is success at this level.
6. Continue to move up to larger sizes until the child succeeds at swallowing a candy of comparable size to a pill (e.g., M&Ms®, Smarties®).
7. At this point, the child should be able to swallow most pills with minimal problems.

Make sure the exercise is fun and relaxed and keep sessions short, so the child does not become tired or stressed. Be sure to encourage the child's successes along the way and be supportive. The child can be encouraged to try different head positions when swallowing, e.g., head tilted back, slightly forward, in the centre, to the left, and the right. Let the child find which head position suits them best.

5.3 Pill Swallowing Techniques

In general, it is best not to throw a tablet or capsule towards the back of the mouth. This is because it can make swallowing more difficult. The following outlines various techniques that can be tried.

Table 9: Guide to Swallowing Pills

• Ask the child to have a drink of water or their favourite drink to moisten their mouth • Place the tablet or capsule into the centre of the child's mouth • Ask the child to take a big sip of their drink, and then swallow
• Yogurts and thick drinks, such as milkshakes, can help ease tablets or capsules down
• Using a straw to drink, with a tablet or capsule already in the mouth, may also help by getting the child to concentrate on the suction of the straw rather than thinking about the tablet or capsule going down
• Put the capsule into a small spoonful of ice cream or a crushed banana. This can help capsules to slip down the throat more easily
The physical properties of capsules may cause them to float in the mouth when taken with water. Some children may wish to try this: • Ask the child look down at the floor instead of up at the ceiling • Slip the capsule into the centre of the child's mouth • Ask the child to take a big sip of their drink while still looking at the floor. The capsule should float to the back of the child's mouth and roll down their throat with the drink
• When children cannot yet swallow whole tablets or capsules, an option could be to empty the contents of a capsule into food or drink • ARVs should not be crushed or compounded into a suspension, as this may affect the stability, bioavailability and dose accuracy the medications.

Adapted From the University of Calgary. How to swallow pills
www.research4kids.ucalgary.ca/pillswallowing

GUIDELINE: PILL SWALLOWING GUIDELINES		
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CHAPTER 6: ANTIRETROVIRAL DRUGS FOR HIV PREVENTION

6.1 Newborn Prophylaxis

6.2 Post-Exposure Prophylaxis for Children <10 years

6.3 Post-Exposure Prophylaxis ARV Regimens for Adolescents 10-19 years

Antiretroviral Drugs for HIV Prevention

6.1 Newborn Prophylaxis

HIV negative mothers should be reminded of safe sex practices, due to the risk of transmission to the healthy newborn, through breastfeeding, if she becomes infected after delivery. The comprehensive management of the HIV Exposed Infant is discussed in further details in Chapter 1.

Children and adolescents who have been exposed to HIV outside of MTCT (e.g sexual assault and needle stick injury) are eligible for post exposure prophylaxis to prevent HIV infection and must be administered within the first 72 hours after exposure.

6.2 Post Exposure Prophylaxis for Children < 10 years

Zidovudine and Lamivudine is the preferred backbone regimen and LPV/r is recommended as the preferred third drug for HIV post-exposure prophylaxis for children aged 10 years and younger (DTG can be used in children whose weight is >20kg). A full 28-day prescription of antiretroviral drugs should be provided for HIV post-exposure prophylaxis following an initial risk assessment.

6.3 Post-exposure prophylaxis ARV regimens for adolescents 10-19 years:

Tenofovir and Lamivudine is recommended as the preferred backbone, and DTG is recommended as the preferred third drug for HIV post-exposure prophylaxis in adolescents. Alternative third drug LPV/r or ATV/r.

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CHAPTER 7: ANTIRETROVIRAL DRUGS FOR HIV TREATMENT

7.1 When to Initiate Treatment

7.2 ARV Regimen for Treatment Initiation

7.3 Monitoring ARV Response and Diagnosing Treatment Failure

7.4 Recommended First and Second Line ARV regimens

Antiretroviral Drugs for HIV Treatment

7.1 When to Initiate Treatment

When to Start ART in Children Younger than 10 Years of Age

ART should be initiated in **ALL** children living with HIV, regardless of CD4 cell count or at any CDC clinical stage.

High-level Urgency of ART Initiation

Rapid initiation of treatment with an expedited discussion of adherence should be done in the following instances:

- a) Any infant with an initial positive virological test result (positive HIV DNA PCR), should commence on ART without delay.
- b) Any child < 24 months of age
- c) Child older than 2 years with CD4 < 25% (or < 350 cells/mm³), especially < 15% (or < 200 cells/mm³)
- d) Children of any age with immunodeficiency or opportunistic illnesses that indicate Class C HIV infection

In other situations, enough time to fully assess and address issues associated with adherence should be allowed before ARV initiation.

Health care providers should monitor the virologic, immunologic, and clinical status of all children with HIV infection, including those who do not initiate ARV.

When to Start ARV in Adolescents (10–19 years of age)

- ARVs should be initiated in **ALL** adolescents living with HIV, regardless of WHO clinical stage and at any CD4 cell count.

- As a priority, ARVs should be initiated in all adolescents with severe or advanced HIV clinical disease (WHO clinical stage 3 or 4) and adolescents with a CD4 count of ≤ 350 cells/mm³.

Timing of ARV in Children and Adolescents with TB

- ARV should be started in all tuberculosis patients living with HIV regardless of CD4 count
- Tuberculosis treatment should be initiated first, followed by ARV as soon as possible within the first 8 weeks of treatment
- HIV-positive tuberculosis patients with profound immunosuppression (e.g., CD4 counts less than 50 cells/mm³) should receive ART within the first two weeks of initiating tuberculosis treatment.
- ART should be started in any child with active tuberculosis disease as soon as possible and within 8 weeks following the initiation of anti-tuberculosis treatment regardless of the CD4 cell count and clinical stage

7.2 ARV Regimen for Treatment Initiation

Table 10: Preferred First Line Regime

	NRTI	PLUS
Neonate up to <3 years	AZT + 3TC	NVP or LPV/r*
Children 3-9 years	ABC +3TC (preferred) AZT +3TC	NVP or EFV**
Adolescent 10-19 years	TDF/TAF +3TC	DTG or EFV

*Preferred for Infants of NNRTI or experienced mothers with poor compliance or who have failed an NNRTI based regime

** Children younger than 3 years should not use EFV

After initiation, ensure close adherence and VL monitoring until undetectable (see monitoring guidelines)

First-Line ART for Neonates and Children up to < 3 years

- For neonates, the NRTI backbone for an ART regimen should be Zidovudine and Lamivudine (AZT + 3TC).
- A LPV/r-based regimen should be used as first-line ART for neonates infected with HIV whose mothers have failed the first-line regime or have a history of poor compliance. Otherwise, Nevirapine is a first-line option. Ensure close viral load monitoring 2 monthly in the first 6 months until undetectable

First-Line ARV for Children

For children, the NRTI backbone should be one of the following, in preferential order

- Abacavir and Lamivudine (ABC + 3TC)
- Zidovudine and Lamivudine (AZT + 3TC)

For children 3 years and older, EFV is the preferred NNRTI for first-line treatment, and NVP is the preferred alternative. EFV is not recommended in children < 3 years.

First-Line ARV for Adolescents

First-line ARV for adolescents should consist of two NRTIs plus an NNRTI or an INSTI:

- TDF/TAF + 3TC + DTG (TLD) is recommended as the preferred option to initiate ARV
- TDF/TAF + 3TC + EFV (TLE) as a fixed-dose combination is recommended as the preferred option to initiate ARV

If preferred regimens are contraindicated or not available, one of the following alternative options is recommended

- ABC + 3TC + EFV
- ABC + 3TC + NVP
- AZT + 3TC + EFV
- AZT + 3TC + NVP
- ABC + 3TC +DTG

For children with TB

Please consult an expert in the management of TB/HIV coinfection. If ARV-naive, start TB therapy immediately and initiate ARVs within 2–8 weeks.

Table 11: ARV Regimen compatible with HIV/TB co-infection

For HIV/TB coinfection on Rifampin	
<3 years	AZT + 3TC+ NVP (increase dose by 40%) * OR AZT+3TC+ twice dose of LPV/r *
>3 years	ABC +3TC +DTG twice daily TDF/TAF + 3TC +DTG twice daily OR ABC +3TC + EFV TDF/TAF + 3TC +EFV

*Revert to standard dosing once the course of Rifampin is completed. Initiate twice-daily Nevirapine without lead-in dosing.

7.3 Monitoring ARV Response and Diagnosing Treatment Failure

- The goal of ARV is to maintain the viral load below undetectable levels. If this is not achieved or maintained, the cause for failure should be identified and modified accordingly.

Definition of Treatment Failure

- Viral load is recommended as the preferred monitoring approach to diagnose and confirm treatment failure.
- Virologic failure is defined by a persistently detectable viral load exceeding 1000 copies/ml (that is, two consecutive viral load measurements within a 3-month interval, with adherence support between measurements) after at least 6 months of starting a new ARV regimen.

Causes of Treatment Failure

- Causes of treatment failure include HIV drug resistance, drug toxicity, and poor adherence to ARV.
- If the expected fall in viral load does not occur (to <400 copies/μl by 2-3 months and < 50 copies/μl by six months) or if the virus re-emerges after being undetectable, non-adherence should immediately be considered and addressed.

Management of Treatment Failure

- Determine lifetime adherence and identify barriers to care.
- Review the patient's lifetime treatment history to determine if potential cross-resistance is present within a given class of drugs. Maternal, partner, or other source case treatment history should be reviewed.
- Antiretrovirals should not be changed unless ongoing poor adherence has been ruled out as the cause of failure, and measures are in place before the initiation of a new regimen. Extraordinary measures such as hospitalization, change of caregiver, or intense psychosocial management, may need to be instituted to ensure adherence.

- Do an intensified 2-month trial of full adherence with aggressive adherence support and repeat the VL at the end of that period
- If treatment failure is suspected, then do resistance testing and consult with HIV specialist
- Select a new regimen based on national guidelines and preferably with reduced pill burden
- The patient should be switched to a new regimen within one month of confirming treatment failure.
- Optimize ongoing adherence support and ARV treatment monitoring

Whenever a regimen is switched:

- Counsel the caregiver or adolescent on reasons for the change in regimen, differences in drug types, dosages, and timing of administration.
- Review with the adolescent and their caregiver the possible side effects of the new regimen.
- Re-assess for social issues that could negatively influence adherence and review the importance of adhering to the clinic visit schedule as well as to the regimen.
- Provide ongoing adherence counselling and support .

Genotype testing

- Detection of the resistant HIV by genotyping is most sensitive if the sample is taken when the patient is adherent but not achieving full virologic suppression.
- Once a drug is discontinued, or the patient is non-adherent, latent drug-sensitive wild type virus (which multiplies faster than drug-resistant virus) takes over as the predominant circulating viral strain. Resistant strains become the minor population.
- When drug exposure resumes, the drug-resistant strains rapidly emerge and will be detected. Therefore, the timing of the sample is critical when interpreting results. If the patient is non-adherent, the drug-resistant strain will not be detected, and failure to detect resistance does not mean resistance is not present. HIV drug resistant samples are to taken after at least 2 weeks of consistent ARV therapy.

7.4 Recommended First and Second Line ARV regimens

Table 12: Sequencing Options for First and Second-Line

	First-line	Second-Line
Infants and Children	AZT + 3TC + NVP AZT+ 3TC + LPV/r ABC (or AZT) + 3TC + EFV	ABC + 3TC + DTG or ABC + 3TC + LPV/r** ABC + 3TC + DTG AZT (or ABC) + 3TC + DTG or AZT (or ABC) + 3TC + LPV/r**
Adolescent	TDF/TAF + 3TC + DTG TDF + 3TC + EFV (or NVP) AZT + 3TC + EFV (or NVP)	AZT + 3TC+ ATV/r or LPV/r AZT + 3TC+ ATV/r or LPV/r TDF/TAF + 3TC+ ATV/r or LPV/r

* DTG dosing approved for children > 20kg

** For Second line use LPV/r if weight < 20 kg Please see formulary

N.B Tenofovir can be considered as an alternative for children >9 years and who weigh >30kg

Second-Line ARV

Second-line ARV for children

- After the failure of the first-line regimen of ABC or TDF/TAF + 3TC the preferred NRTI backbone option for second-line ARV is AZT + 3TC
- After the failure of a first-line regimen containing AZT + 3TC the preferred NRTI backbone option for second-line ARV is ABC or TAF + 3TC
- After the failure of a first-line NNRTI-(Nevirapine/Efavirenz) based regimen, children > 20 kg should be switched to DTG
- If the child's weight is < 20 kg the child should be switched to LPV/r.
- After the failure of a first-line LPV/r-based regimen, children > 20 kg should be switched to DTG.

Third-line ARV

Do HIV drug resistance testing and refer and discuss with Key HIV Physicians.

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CHAPTER 8: HIV INFECTED ADOLESCENTS

- 8.1 Perinatally-Acquired HIV Infected Adolescents (PHIVA)**
- 8.2 Adolescents Living with HIV (ALHIV)**
- 8.3 Seek, Test, Treat, Retain, Suppress**
- 8.4 Comprehensive Care for ALHIV**
- 8.5 Counselling**
- 8.6 Sexual and Reproductive Health**
- 8.7 Psychosocial Care**
- 8.8 Mental Health**
- 8.9 Collaborative HIV Prevention and Adolescent Mental Health Programme (CHAMP+)**
- 8.10 Transition to Adult Care**

HIV Infected Adolescents

Adolescents and young adults are at increased risk for HIV due to the many developmental, psychological, social, and structural transitions that converge in this period of the lifespan.

Table 13: Age Ranges According to WHO

Group	Age range (according to WHO)
Adolescents	10–19 years
Youth	15–24 years
Young people	10–24 years

The goals of comprehensive HIV care are to:

- Reduce HIV-related illness and death
- Improve the quality of life
- Improve the lives of families and communities affected by HIV
- Prevent the further spread of HIV

There are 2 populations of adolescents living with HIV. There are those who acquired HIV through perinatal transmission and require HIV care from the time of their birth. Then there are those who acquired HIV through risk behaviours in adolescence, such as sexual transmission or injection drug use (the latter is uncommon in Jamaica).

8.1 Perinatally-Acquired HIV Infected Adolescents (PHIVA)

Adolescents with perinatally acquired HIV generally display the following characteristics:

- Have typically been in care since they were young. However, some adolescents may be diagnosed later in life
- Likely began their experience in HIV care and treatment when they were children
- Have typically been on ARV for many years; may even be on a 2nd or 3rd line regimen and experience medication fatigue
- Often look young for their age, due to delays in development and overprotection by caregivers; and can often have inadequate social skills

8.2 Adolescents Living with HIV (ALHIV)

- In Jamaica, HIV testing is disproportionately higher among girls due to ANC. However, HIV prevalence estimates among male and female adolescents are the same and was found to be high among young MSM based on a UNICEF assessment in 2015
- They may be socially experienced, possibly more so than many of their peers
- They may be relatively inexperienced in terms of navigating the health care system and dealing with health workers
- Are typically treated as adults, with their treatment directed by adult guidelines
- Among the HIV adolescents infected through sexual assault, there is an intensified psychological impact compared with other methods of transmission
- High-risk behaviours such as inconsistent condom use, drug use (alcohol, cigarettes, marijuana) increase their risk for ongoing sexual transmission
- Multiple factors affect treatment adherence

Both populations of HIV seropositive youth require tailored approaches for their care to be successful. The approach for all adolescents should be family-centered and developmentally appropriate.

While the components of the adolescent package of HIV care closely resemble those of the adult package, the way these components are delivered has an important impact on their uptake and success among adolescents.

The cascade of care for individuals living with HIV begins with diagnosis, counselling, linkage to care, and initiation of antiretroviral therapy (ARV). However, the work of patients and providers has only just begun.

Strict adherence to ARV, persistent adherence to ARV, and retention in care are necessary for viral suppression and require collaboration of patients, health care staff, social workers, and physicians.

To be effective, adolescent services must:

- Be integrated or available at one facility (1 stop shop)
- Be age and developmentally appropriate
- Be responsive to the needs of both perinatally infected adolescents and those infected later in childhood or adolescence
- Youth-friendly
- Family-focused
- Have stigma free, supportive services

8.3 Seek, Test, Treat, Retain, Suppress

In Jamaica, there is an increased prevalence of HIV in 15-19-year olds, which increases up to age 24 years.

The WHO strategies for health interventions for the adolescent include the following:

1. In generalized epidemics, HIV testing and counselling with linkage to prevention, treatment, and care is recommended for all adolescents.

2. Counsel adolescents with HIV about the potential benefits and risks of disclosure of their HIV status to others and empower and support them to determine if, when, how, and to whom to disclose.
3. Promote adolescent HIV treatment, care, adherence and retention in care through:
 - community based service delivery
 - training for health workers
 - interventions to support onward disclosure
 - improvement of adolescent treatment literacy and mental health
4. Initiate lifelong ARV to all adolescents living with HIV, regardless of CD4 cell count and at any WHO clinical stage.
5. Establish linkage and referral pathways to ensure a comprehensive continuum of care, including the transition from paediatric/adolescent to adult HIV services.
6. Address the needs and vulnerabilities of adolescents from key populations, including drug users, people who sell sex, males who have sex with males, and transgender people.
7. Address legal and policy constraints, service coverage, and barriers to access. In Jamaica, the Law Reform (Age of Majority) Act, Section 8, states that adolescents 16 years old and older may access and consent to medical care without the presence or consent of a caregiver or parent.

“Where a minor has attained the age of sixteen years his consent in respect of any surgical, medical or dental treatment to himself shall be as effective as it would have been if he were of full age; and where a minor treatment has, by virtue of the provisions of this section given an effective consent in respect of any such treatment, it shall not be necessary for consent to be obtained from the parent or guardian of that minor in respect of that treatment”

8.4 Comprehensive Care for ALHIV

Comprehensive care for ALHIV includes the provision of the services listed in the clinical assessment checklists in the following table.

Table 14: Steps in Comprehensive Care for ALHIV

✓	Steps
	1. Take history
	Take a complete medical and social history, including prenatal, birth, and family history
	Confirm HIV infection status
	Identify concomitant medical conditions (e.g., TB disease, hepatitis B or C infection, other co-infections or OIs, pregnancy in adolescent girls)
	Enquire about disclosure to the adolescent (if perinatally infected, take time alone with the caregiver to discuss) or disclosure to others
	Enquire about HIV and treatment status of the family and household members
	Enquire about concomitant medication (e.g., CTX, oral contraceptives, traditional therapies)
	Review Immunization status
	If clinically indicated, undertake a nutritional status assessment
	- Ask about sexual activity including sexual abuse/ statutory rape (alone with adolescent) - Enquire about condom and other contraceptive use
	Conduct psychosocial assessment and provide counselling, referrals, and support (see <i>Appendix 3 HEADSS Interview Questions</i>)
	Assess any other practical needs, such as legal support, housing, school/ career, and financial
	2. Conduct physical exam
	Assess growth and nutrition (weight, height, and BMI), as appropriate for the age
	Assess development and neurodevelopment, as appropriate for age

✓	Steps
	• Conduct physical examination, including Tanner staging
	• Conduct skin exam (tattoos, bruises, acne) and scoliosis evaluation
	• Screen for STIs in adolescents who are sexually active
	• Screen for pregnancy in sexually active adolescent females
	• Screen for TB; screen for other OIs and other concomitant conditions
	• Discuss findings from physical examination with ALHIV and his or her caregivers
	3. Make a laboratory assessment plan (see laboratory evaluation)
	• CD4 count (All HIV positive patients must have an initial CD4 count followed by 3 and 6 monthly tests until the patient is virally suppressed). • Viral load assessment is to be performed every 6 months if unsuppressed and yearly if suppressed. • CBC (Hb, WBC, differential, platelets) • Urea and electrolytes, fasting glucose, fasting lipid profile • Liver enzymes, serum proteins • Mantoux test • Hepatitis B
	4. Make assessments
	• Review findings from history, physical assessment, and laboratory work and make the diagnosis
	• Assess the CDC/WHO clinical stage. If on ARV, determine if there are any new events
	• If not on ARV, prepare for ARV
	• Decide if Bactrim prophylaxis is necessary
	5. Make decisions
	• Discuss the prevention of illnesses (OIs, including TB, STIs, diarrhea and other illnesses) and initiation or continuation of Bactrim, and any other medications
	• If applicable, discuss prevention of STIs, positive prevention, and prevention of unintended pregnancy; provide condoms and contraceptive counselling and methods
	• Discuss the treatment of current illnesses identified in physical examination
	• Discuss adherence and side effects

✓	Steps
	• If applicable, provide nutritional counselling and support
	• Provide counselling, support, and referrals based on psychosocial assessment and needs • Institute urgent interventions and collaborations with the Child Development Agency, Center for Investigation of Sexual Offences and Child Abuse for the appropriate investigations and medicolegal investigations of child sexual abuse/ statutory rape if suspected
	6. Agree on an action plan
	• Agree on key action steps from history and physical examination
	• Discuss when to seek medical care; for example, with an unexpected illness or side effects
	• Reiterate agreed upon plan to support adherence to medications
	• Discuss steps to live positively and prevent further HIV infections
	• Agree on key action steps based on psychosocial assessment (e.g., reduce alcohol intake, discuss HIV-status with a friend, join a support group)
	• Provide referrals, including the name of person/agency, address, and contact information of referral point. If possible, make the appointment on behalf of ALHIV
	• Schedule next visit as per national guidelines: • Schedule an early appointment within 2 weeks of ARV initiation to monitor adherence, if clinically unwell or if the CD4 is declining • Monitor 2 weekly and monthly as necessary or until no adherence issues are detected • Review 2 monthly until virologically suppressed, then every 3 months • Encourage ALHIV to drop in (without an appointment) if a problem arises and to participate in other clinic activities, such as support groups

There should be proper documentation, in the medical records, of the steps carried out in the comprehensive care provided to the ALHIV. All necessary information should also be entered in the HIV database (Treatment Site Information System 2-TSIS2). Monthly cascades should also be generated at each treatment site, using TSIS2, to report on the retention, adherence and viral suppression of the ALHIV clients.

8.5 Counselling

How to Communicate with an Adolescent

- Establishing a comfortable and open relationship using the listening and learning skills discussed in Appendix 11. This is the foundation for good communication
- Encourage the adolescent to talk about general things that interest him or her before moving onto specific issues in the client's life. (For example, *"Did you hear about the football match last night? How is school going? I like the blouse you're wearing; did you sew it yourself?"*).
- Engage and take an interest in the adolescent as a person, and not just in his or her physical condition.
- Explain confidentiality and note that there are some situations in which it may be necessary to breach confidentiality.
- Do not ask too many questions, and listen attentively
- Use an interactive, participatory style of communicating and give the adolescent enough time to explain his or her ideas and decisions to you.

Table 15: Do's and Dont's of Counselling

Counselling includes:	Counselling does NOT include:
• Establishing supportive relationships • Having conversations with a purpose (not just chatting) • Listening carefully • Helping people tell their stories without fear of stigma or judgement • Giving correct and appropriate information • Helping people make informed decisions	• Solving another person's problems • Telling another person what to do • Making decisions for another person • Blaming another person • Interrogating or questioning another person • Judging another person

Table 15: Do's and Dont's of Counselling	
Counselling includes: • Exploring options and alternatives • Helping people to recognize and build on their strengths • Helping people develop a positive attitude toward life and to become more confident • Respecting everyone's needs, values, culture, religion, and lifestyle • Being willing to trust clients' feelings and decisions, which may be the right ones for them at that time, given their situation	Counselling does NOT include: • Preaching to or lecturing another person • Making promises that cannot be kept • Imposing one's own beliefs on another person • Providing inaccurate information

8.6 Sexual and Reproductive Health

Risk Mitigation

Unsafe sex is the primary means by which HIV is contracted and spread among adolescents and youth. It is very important for health workers to be comfortable to have honest, factual discussions about sex, sexuality, and reproduction with their adolescent clients.

These discussions can provide adolescents with the information they need to protect themselves and their partners from STIs and unplanned pregnancy. Exploring HIV transmission risk from sexual abuse should be included.

Recognizing sexual abuse can be difficult as there is usually no physical evidence that an adolescent has been sexually abused. Physical signs are more prominent in children. The adolescent may not volunteer information as they are ashamed, so the health care worker (HCW) must be alert. When adolescents report that they are being or have been sexually abused, there is a high probability that they are telling the truth.

Changes in the adolescent's behaviour such as poor school performance, depression, promiscuity, drug abuse, delinquency, self-destructive behaviours, unexplained gifts, money, or interactions online are possible clues.

When discussing sexual abuse with adolescents, the focus of the discussion must be on stopping the abuse, counselling the victim, identifying ways to support healing and evaluating and treating STIs with appropriate multidisciplinary team approach including referrals to social and mental health services and Centre for the Investigation of Sexual Offences and Child Abuse (CISOCA), an arm of the Jamaica Constabulary Force.

Follow national testing guidelines when providing HIV counselling and testing to adolescents of unknown HIV-status who have experienced sexual abuse.

Table 16: Sexual and Reproductive Health Questions for the Client.

✓	Questions for the client:
	1. Is the client sexually active?
✓	<i>Some adolescents have sex with their partners.</i>
✓	<i>Are you having sex?</i>
	2. If yes, with whom?
✓	<i>Are you having sex with males, females, or both?</i>
✓	<i>How many partners do you have right now? How many partners have you had in the past year?</i>
✓	<i>What is the HIV-status of your partner(s)?</i>
✓	<i>Does your partner know you have HIV?</i>
	3. What are the client's sexual practices?
✓	<i>Do you have vaginal sex? Oral sex? Anal sex?</i>
✓	<i>What family planning method did you use the last time you had sex?</i>
✓	<i>When was the last time you used a condom?</i>
✓	<i>Has anyone caused you harm in the past, for example, hurt you physically or made you have an unwanted sexual encounter?</i>
✓	<i>Have you ever used cigarettes, alcohol, or other drugs? If so, how often in the last week have you used cigarettes, alcohol, or other drugs?</i>

Sexual Orientation

- An adolescent who realizes that he or she may be gay, bisexual, or transgender may feel isolated and depressed.

- The health worker does not have to be an expert on sexual orientation.
- The most important thing is that the health worker is willing to listen to adolescent clients in a non-judgmental way and provide them with any necessary referrals.
- It is important that the HCW not project their own opinions or values about sexual behaviours as adolescents will be less willing to be open and honest about their feelings. They should always be made to feel comfortable talking about their sexual concerns, questions, and behaviours.
- If health workers do not talk about sex and sexual behaviours with clients, they may not get the information, skills, and supplies they need to protect themselves and their partners and to reduce their risk of HIV, STIs, sexual violence, discrimination, and unplanned pregnancy.

Safer Sex Counselling

HIV is transmitted from one person to another through four (4) body fluids

- semen,
- vaginal secretions,
- blood, and
- breast milk.

Any activity during which one or more of these body fluids is passed from one person to another could pose a theoretical risk of HIV transmission if;

- 1) the body fluid is from a person infected with HIV, or
- 2) if the body fluid enters the bloodstream of another person.

The HCW should encourage adolescents to abstain from activities that are high risk involving these exchange of body fluids.

If adolescents are already sexually active, studies show that an abstinence-only strategy is ineffective. Encouraging the substitution of high-risk activities with no risk or low-risk activities will be a more effective and achievable goal.

Counselling Adolescents on Having Children in the Future

Many ALHIV have questions about whether they can safely have children in the future.

It is important that ALHIV know the facts about pregnancy and preventing mother-to-child transmission BEFORE they become pregnant.

The adolescent should be advised that it is safest to wait until adulthood to become pregnant and have children. There are many health-related, psychological, social, and economic risks associated with having a baby during adolescence.

The safest time to get pregnant is when the woman or male partner with HIV:

- Is taking and adhering to their ARV regimen,
- Has a CD4 cell count above 500 and has an undetectable viral load
- Is healthy, i.e., they do not have any opportunistic infections (including TB) or advanced AIDS, and
- Is taking and adhering to their ARV regimen

Adolescent clients should be encouraged to talk with health workers about pregnancy and PMTCT if they are thinking of having children. Health workers should also encourage the partners of these adolescent clients to come to the clinic for education and counselling on these topics.

Contraception

Ensuring integration of HIV prevention services into reproductive health, treatment care and support programmes, increase the uptake of family planning.

A “one stop shop” provides a friendly space where young people can access information and increases uptake to Adolescent Sexual and reproductive Health (ASRH) services.

If the services are not available on site, they should be accessible nearby.

The key message is dual method use, using condoms and a contraceptive method for good protection from infection and unintended pregnancy.

Table 17: Special Considerations for Adolescent Clients

Birth Control	Types	Comment
Barrier Methods	Male Condom Female Condom Diaphragm Sponge	• Adolescents require skill development and practice to learn how to use condoms and to negotiate their use with sexual partner(s). • Adolescent girls are frequently not assertive about the use of condoms if their partner rejects the idea — they require counselling and peer support to feel empowered to negotiate condom use • Failure rates (i.e., unintended pregnancy) for condoms are high, especially among adolescents who often do not use them consistently or correctly. • Females should use an additional contraceptive method to prevent pregnancy.
Short-Acting Hormonal Methods	Oral Contraceptive Pills (OCP)	• Should always be used in combination with condoms to provide STI/HIV protection. • Rifampicin lowers the hormonal levels of OCP, patch, and vaginal ring. Recommend dual therapy with condom, diaphragm or IUD while taking rifampin

Birth Control	Types	Comment
Short-Acting Hormonal Methods Cont'd	Oral Contraceptive Pills (OCP) Cont'd	• ARVs may adversely affect the efficacy of low-dose OCP and/or increase their side effects. • Women taking ARVs who want to use OCP can be given a formulation with at least 30mcg of estrogen. • They should be counselled about the importance of taking OCP on time every day (without missing pills) and about consistent condom use.
	Emergency Contraceptive Pill (ECP)	• Adolescents receiving ECP should be counseled on adopting a regular contraceptive method and on condom use for dual protection. • ECP should be taken as soon as possible after unprotected sex. • If more than 72 hours have passed since unprotected sex, the client should not be given ECP. If the client vomits within 2 hours of taking a dose, the dose should be repeated. • If she does not get her period within one week of when she expected it, she should return to the clinic because she could be pregnant.

Birth Control	Types	Comment
	Patch and Vaginal Rings	Patients with contraindication to oestrogen i.e., Systemic Lupus, history of thrombosis, risk factors for cardiovascular disease, and hypertension, should not use the OCP, patch or vaginal ring.
Long-Acting Reversible Contraceptives	Implantable Rod (e.g., Norplant)	Should always be used in combination with condoms to provide STI/HIV protection.
	Intrauterine Device	- IUDs are appropriate for adolescents in stable, mutually monogamous relationships. - Careful screening for STIs before insertion is critical, and IUDs are not recommended for ALHIV with advanced HIV disease or AIDS (especially if they are not on ARV). - Encourage adolescents choosing an IUD to use condoms for dual protection. - May be used in patients with contraindication to OCP, patch, and rings (i.e., cardiovascular disease or hypertension).
	Progestin Shots (e.g., Depo-Provera)	Should always be used in combination with condoms to provide STI/HIV protection.

8.7 Psychosocial Care

Psychosocial support addresses the ongoing emotional, social, and spiritual concerns and needs of adolescents, the family, as well as caretakers of children living with HIV.

- All adolescents need support coping with normal developmental issues, such as wanting to feel normal and accepted, and wanting to fit in with peers.
- In addition to the psychosocial needs and challenges that all adolescents face, PHIVA and ALHIV may also experience HIV-related stressors and additional vulnerabilities and challenges.
- Children who lose their parents often internalize their psychological turmoil and feel the negative effects of parental death up to 2 years afterward.
- Lesbian, gay, bisexual, transgender, queer youth are susceptible to emotional distress, psychiatric morbidity, multiple disparities, stigma, abuse, violence, and isolation
- If a health worker thinks an adolescent client has serious issues that threaten his or her life or immediate safety, these issues must be addressed **IMMEDIATELY**.
- An important part of helping adolescents cope with issues is encouraging their caregivers to strengthen their relationship with them.
- Peer support is an important source of psychosocial support for HIV infected adolescents.
- Health workers have a role in initiating and facilitating peer support groups — and in linking adolescents with existing support groups.
- It is important to provide a welcoming and confidential environment.
- Case conferences should be held regularly to discuss challenges and strategies to help families cope.

8.8 Mental Health

All adolescents should be screened for mental health disorders and substance use disorders. In the clinic, a psychosocial screening tool (see **Appendix 4 & 5**) can be administered by the adherence counselor or other members of the treatment care and support team and facilitate an early referral.

A psychologist needs to be a part of the treatment care and support team for evaluation and psychosocial and behaviour management therapies.

Additional screening for depression and childhood or adolescent anxiety would be useful to identify at-risk children or adolescents.

8.9 The Use of the Collaborative HIV Prevention and Adolescent Mental Health Programme (CHAMP+) to Plan for Disclosure and Transition to Adult Care

HIV infection is a serious threat to the well-being and mental health of children and adolescents. Even as the rate of HIV/AIDS deaths decline, the youth account for an estimated 45% of new infections worldwide. The Collaborative HIV Prevention and Adolescent Mental Health Programme (CHAMP+), is an adaptation of the CHAMP primary model that was developed to intervene at the family level, through parent/child sessions, to enable safe and healthy communication on HIV knowledge and direct decision making for HIV prevention activities. CHAMP+ focuses on HIV and the impact on the family, HIV health education, dealing with stigma and discrimination, HIV medication adherence, as well as empowering youth in their decision making related to sexual behaviour and disclosure.

In Jamaica, CHAMP+ has been modified for the local context and was used as the foundation for CHAMP++. CHAMP++, is used to reinforce existing structures and expand mental health services for adolescents and youth, through family based interventions

delivered through multi-level group modalities. These activities are spearheaded by psychologists, social workers, adherence counsellors and community health aides, who are able to provide family support and channels for open communication. CHAMP++, sets the foundation for disclosure of HIV status to children and adolescents, provides a forum for empowering the adolescent to manage their health and encourage medication adherence, and enables a smooth transition from paediatric/adolescent care to adult care.

8.10 Transition to Adult Care

A successful transition involves a client-centered process and a developmental approach, it is not a one-time event. In Jamaica, adolescents who transitioned prematurely to adult care had a reduction in the prevalence of viral suppression compared with their counterparts who had not been transferred.

Taking on a greater role in self-care and self-advocacy, may be challenging for adolescents, depending on their level of development and maturity.

Not all ALHIV will be ready to make the transfer to adult care at the same age. Health workers must consider their cognitive and physical development, their emotional maturity, their support at home and in the community, and their health status. Critical linkages with the adult centered team is important for continuity of care

Goal of Transition

The goal of transition is to ensure the provision of uninterrupted, coordinated, developmentally and age-appropriate, and comprehensive care before, during, and after the transition.

Transition is applicable to every ALHIV as they mature into adulthood. All adolescents require support both within and outside of the clinic setting to take greater ownership over their health care, behavior, lives, and adherence to care and treatment.

The transition to adult care generally occurs in parallel with an adolescent's emotional and physical maturation into adulthood. The effective transition must allow for the fact that adolescents are undergoing changes that impact much more than just their clinical care. Adolescents' psychological maturation may be influenced by how and when they assume responsibility for their care and vice versa.

Note: Not every adolescent will be able to reach 100% independence from their caregivers. This is particularly true for adolescents who have moderate or severe developmental delays. In such cases, caregivers will likely need to stay involved in the adolescent's care after the transition to the adult clinic. Pediatric/adolescent and adult clinics will need to consider special arrangements to accommodate developmentally or otherwise disabled clients.

Ideally, adolescents should be able to do the following before transitioning:

- Make, cancel, and reschedule appointments
- Arrive to appointments on time
- Call ahead of time to schedule urgent visits
- Request prescription refills correctly and allow enough time for refills to be processed before medications run out.
- Recognize when to seek medical care for symptoms or emergencies
- Identify symptoms and describe them
- Negotiate multiple providers and different types of clinic visits
- Establish a good working relationship with a case manager at the pediatric clinic, which will enable them to work effectively with the case manager at the adult ARV clinic
- Ask questions and ask for help when needed
- Understand and be able to describe their disease, care and treatment plan, including the medicines they are taking
- Get the results of every test and understand the results
- Join a support group if available
- Follow up on all referrals

Table 18: Transition Preparation Timeline

10–12 years old	13–16 years old	17–19 years old
• Encourage caregivers to disclose to the child fully • Solicit direct conversation with the adolescent • Increase one-to-one meetings and counselling sessions with the adolescent • Begin to explain medications and adherence • Deal with early adherence issues and challenges • Link adolescents to support groups	• Assist adolescents with a calendar for appointments and medicines • Ensure adolescents understands diagnosis, needed medications, adherence, health precautions, positive living, and positive prevention • Teach adolescents to identify symptoms of sadness and a person they can speak to when sad • Educate concerning stigma and its effects • Assist with understanding of CD4 and Viral Loads levels	• Ensure responsibility in making and keeping appointments • Provide ALHIV with copies of medical records and any other forms of documents required by the adult clinic • Review medical history with the client • Encourage questions about adolescent's care plan, treatment regimen, and possible changes • Transfer medical records to the new provider, highlight key issues • Visit the adult clinic with the adolescent client

Source ICAP. Adapted from: AETC National Resource Center; New York/New Jersey AETC; Texas/Oklahoma AETC and Florida/Caribbean AETC. (2003). *The HIV perinatally-infected adolescent: A developmental approach, Practitioner transition checklist, and timeline*. Available at: <http://www.aids-ed.org/aidsetc?page=etres-display&resource=etres-272>

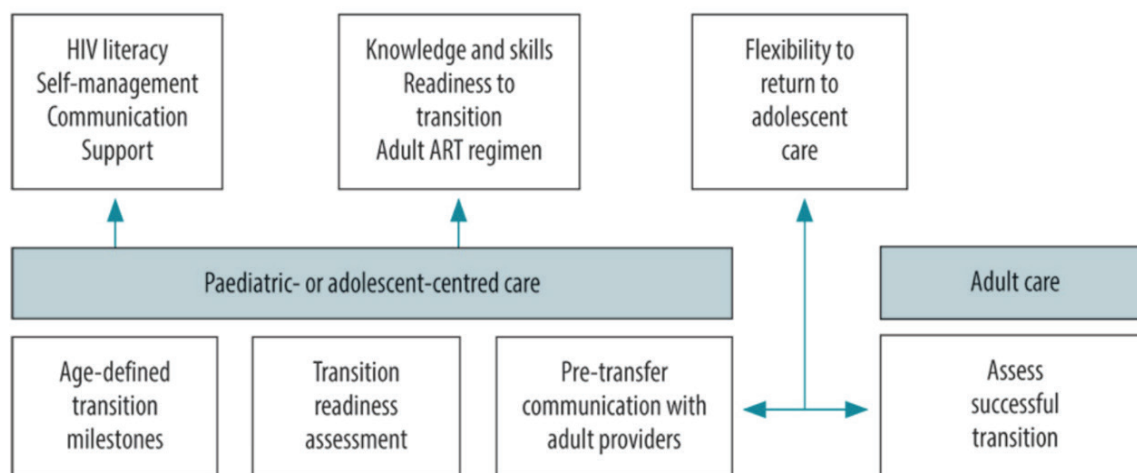
Table 19: Key Elements and Definitions of Successful Transition of Children and Adolescents to Adult HIV Care.

Readiness Assessment	Adolescents are assessed by a healthcare worker at 17 years of age ensuring: • transition knowledge and skill milestones; (HIV literacy; self-management; communication and support); • optimized regimen of antiretroviral therapy (on an oral adult regimen); • no new opportunistic infections or active infections • school, social and family concerns are addressed • Understanding of the transition process.
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Caregiver Readiness	Caregivers are willing to allow adolescents to take responsibility for their own health.
Communication with Adult Clinic	Adult Clinic staff are aware of the adolescents being transitioned to their clinic.
Peer Support	Support groups are available where possible.
Special Considerations	Transition process can be slower or faster for pregnant adolescents, those with delayed or early milestones and adolescents with disability.

Bulletin of the World Health Organization 2019; 97:837-845. doi: <http://dx.doi.org/10.2471/BLT.19.232702>

Framework for the Transition of Children and Adolescents to Adult HIV Care



Bulletin of the World Health Organization 2019; 97:837-845. doi: <http://dx.doi.org/10.2471/BLT.19.232702>

Post-transition follow-up is recommended to include contact at 3 months to evaluate linkage to care and at 6–12 months to assess retention and viral suppression.

GUIDELINE: HIV INFECTED ADOLESCENTS		
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CHAPTER 9: DISCLOSURE

9.1 Stages of Disclosure

9.2 Planning and Preparing for Disclosure by Age

9.3 Key Messages for Disclosure

Disclosure

Caregivers fear negative consequences from disclosure, such as psychological problems, inability to comprehend and deal with the diagnosis and stigma, and unintended disclosure to others.

The reality from published studies shows that disclosure tends to have positive long-term impacts for all invested individuals: the child, the parents, and the family:

- most children who have been told their HIV diagnosis do not experience long-term negative effects with a large percentage of children in one study reporting disclosure as a positive event for them
- children who know their HIV status have higher self-esteem compared to children who are unaware of their status
- disclosure of HIV status may support the uptake of, and adherence to ARV
- disclosure to children and adolescents may present an opportunity for increased access to adherence or psychosocial support (PSS)
- Parents who have disclosed the status to their child experience less depression compared with those who did not

World Health Organization (WHO) guidelines recommend that:

- children of school age are told of their HIV-positive status;
- younger children should be informed incrementally to accommodate their cognitive skills and emotional maturity, in preparation for full disclosure at an older age.

9.1 Stages of Disclosure

The stages of the disclosure process include:

- **Pre-disclosure:** The child has not yet been told about their HIV status.

- **Partial disclosure:** Telling the child the truth, but not the whole truth, usually withholding the name of HIV (can be a step in the full disclosure process).
- **Full disclosure:** This stage provides the child or adolescent with the name of the diagnosis and full information and knowledge about HIV.
- **Post-disclosure:** Period following disclosure, commonly described as early post-disclosure (first three to six months) and late post-disclosure (over six months) periods.

9.2 Planning and Preparing for Disclosure by Age

Disclosure in Children and Adolescents

There is a difference in HIV disclosure for children with vertically versus horizontally acquired HIV.

For the vertically HIV infected child, the diagnosis is known. Caregivers may choose to delay or postpone the disclosure process out of fear of stigma and being identified as a source of HIV infection and therefore feel guilty and have the fear of being blamed, or rejected.

For the horizontally infected adolescents, they require counselling on disclosing their status at home and to their peers. They require strong social support which facilitates retention in care and adherence to treatment.

Disclosure to Caregivers of Displaced Children

Children may be assigned to a new primary caregiver for a variety of reasons. Once it is established who the new primary caregiver is, there should be full disclosure of the child's status and education of the caregiver to facilitate comprehensive treatment and viral suppression. Discuss with whom it is safe to disclose the child's status to ensure confidentiality within the child's or adolescent's new home environment.

Within the institutionalized setting, there should be disclosure and confidentiality among caregivers who are trained to administer ARVs in a supportive environment. The importance of confidentiality should be stressed, as children/adolescents may be the subject of bullying and discrimination by other wards of the state.

If there is no disclosure to a caregiver administering medication due to parent refusal, adherence will be compromised. Intensive family support, assisting the parent with disclosure or identification of alternate caregivers are strategies that can be used.

Caregivers and all children from the age of 3 years should start being prepared for disclosure as follows:

- Non-Disclosure (under 2 years)
- Partial Disclosure (3-9 years)
- Full Disclosure (from 10 years, usually by entry to high school years), legal and ethical principles

Assess Readiness for Disclosure

HCWs should build up trust and establish rapport with children and adolescents and their caregivers, from the start of providing services. It is important for a HCW to discuss the following points with the caregivers:

- Ask what the child or adolescent has been told about their health and medications.
- Highlight the benefits of disclosure.
- Inquire about potential barriers to disclosure.
- Offer support and **agree on a plan** to disclose the HIV status.
- Prepare the caregiver for handling questions from the child or adolescent.
- For horizontally infected adolescents and youth, discuss the need for a support network at home and evaluate the potential family members/partners for disclosure

If the HCW finds that there is caregiver reluctance to disclose, some approaches include:

- Respecting the caregivers' reasons for fearing or resisting disclosure and attempting to understand the factors associated with their reluctance
- Working with the caregivers to develop a plan that addresses their concerns and develop the timeline for future disclosure
- Caregivers can be referred for further counselling if required

Throughout the phases of disclosure (Pre, partial, full, and post-disclosure), assess the child's or adolescent's emotional well-being and functioning in the following areas:

- School functioning
- Family and peer relationships and support
- Interests and activities in school and within the community
- Mood and behaviour. (e.g., symptoms of mood swings, depression)

General Principles for Disclosure

- The disclosure process should not be rushed, but the timing of disclosure becomes more urgent as the child nears adolescence and for adolescents in need of support services
- Use of clear and developmentally appropriate explanations of the disease and diagnosis is mandatory. For children or adolescents with neurocognitive impairment, disclosure is independent of age and dependent on the level of cognitive function
- Date of disclosure should not coincide with other important life events such as birthdays, holidays, graduation, etc.
- Disclosure should promote the sharing of feelings, but also should accept silence
- HCWs should encourage the child or adolescent to ask questions during the disclosure process
- When available, HCWs should use developmentally appropriate educational materials such as digital illustrations or simple drawings

Table 20: Readiness Assessment Criteria

Readiness Assessment Criteria: Partial and Full Disclosures to a Child Aged 5-10 Years	Check Mark
Confirm with the caregiver that they agree to initiate disclosure	
Determine the caregiver's knowledge about HIV and discuss the benefits as well as the risks of disclosure	
Assess what the caregiver has discussed, if anything, with the child regarding his or her HIV status or medications to date	
Ascertain whether there are other people living with HIV in the child's household	
Evaluate the child's daily functioning in school	
Identify whether significant events took place in the past six months, such as a recent family death, divorce, separation, loss of a friend, or distant relocation	
Assess available family and/or community support for the caregiver and the child	
If the mother is the caregiver, discuss any of her feelings of guilt and help her to address them	
Assessment with the Child and Caregiver	
Assess the information the child has about the medicines he or she has been taking	
Determine the information the child has about his or her health and HIV status	
Assess whether the child has difficulties in building and maintaining friendships with peers at school and/or at home with siblings	
Evaluate the child's daily mood at home and at school	

9.3 Key Messages

To initiate partial disclosure, a child or adolescent living with HIV needs to understand why they attend the clinic and takes medications. HCWs and caregivers can start by:

- Explaining to the child that there are viruses (can be referred to as germs) in the blood that can weaken the immune system (soldiers in the body that fight germs). For the older child or adolescent, explain the concept of “good cells” (CD4 cells) and viruses (viral load).
- Explaining that the medicines, antiretroviral (ARV) medications, help put germs to sleep and help them to stay healthy.
- Using simple drawings or when possible (no artistic skills required), printed or digital aids to explain the concept of immune cells being attacked by the virus and medications stopping them.
- Indicating blood tests are needed every few months to check on the number of the body’s “good cells.”
- Discussing the importance of taking medicines every day and staying healthy.

For full disclosure:

- Build on the available knowledge about chronic illness and proceed with full disclosure
- Ask: What have you heard about HIV? (Or what do you know about HIV?)
- Explain that HIV in the blood can weaken the immune system (soldiers in the body that fight germs)
- The real name of the germ is HIV – (Correct any misinformation) – naming HIV and explaining how it can be treated
- Discuss modes of transmission and explain that the transmission was not anyone’s fault

Following full disclosure:

- Provide concrete examples of living positively with HIV. For example: provide anonymous stories of older children with undetectable viral loads who are doing well.
- Explain that there are many children also taking medicine as a normal part of their day and they are just like other children who don't have HIV
- Openly discuss the stigma and its causes (ignorance, prejudice and fear) and provide empowering messages.
- Discuss issues of protecting the confidentiality of personal information and HIV status within the family and the community. For example, "You should only discuss your HIV status with their parents/caregivers. If you want to tell someone else, speak with your family first and decide together."

Throughout and after disclosure:

- Pause periodically, and ask the child what they are feeling at the moment, without rushing answers; accept silence.
- Observe the caregiver's emotional response and provide support and reassurance.

Wrapping up:

Pause to observe the child/ adolescent's immediate reaction and be ready to support an emotional response with words of reassurance.

During the wrap up session:

- Provide encouragement, reassurance, and encourage the caregiver to be supportive of the child at home
- Advise the child, adolescent, and caregiver to talk more at home and write up any questions that arise so they can discuss them at their next visit.

- Remind the child, adolescent, and caregiver, of the benefits of disclosure, as discussed earlier
- Among older adolescents, discuss prevention of HIV transmission to a sexual partner and sexuality. Provide examples of positive parenthood, partnerships, and marriage.
- Provide enhanced adherence counselling reinforcing messages about the benefits of taking medications, seeing a medical provider regularly, and living a healthy lifestyle.
- If the child/ adolescent requires any referral, such as psychosocial counselling and peer support, provide referrals with clear instructions.
- End the session by providing any available printed information to take home and showing respect to the child and the caregiver.

Post-Disclosure Evaluation and Follow-Up

Like adults, children may go through a period of denial, anger, or self-pity following disclosure of their HIV status. Once the HIV status has been disclosed to the child or adolescent, there should be monitoring and follow-up in the short and long term, providing support, additional information, and evaluating for any adverse outcomes. The purpose of the follow up is to:

- Assess positive (such as improved self-confidence, self-awareness) and any negative outcomes (such as stigma and depression) in both the caregiver and the child or adolescent
- Review the understanding of the child or adolescent about their HIV status
- Monitor how well they are coping with the diagnosis and treatment disease
- Identify needs for any further referrals and peer support

Disclosure of HIV status is not a one-time event, but rather a process involving ongoing discussions about the disease as the child or adolescent matures cognitively, socially, emotionally, and sexually.

GUIDELINE: DISCLOSURE		
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CHAPTER 10: ADHERENCE

Adherence

Youth living with HIV are at risk of poor adherence to ARV because they are in the midst of a developmental period characterized by less inhibition, increased risk-taking, and decreased parental support and oversight.

Perinatally and behaviourally HIV infected adolescents both experience barriers to care. PHIVA report significantly more barriers than ALHIV, the most common being forgetting, not feeling like taking medication due to medication fatigue, and not wanting to be reminded of HIV infection. A review of over 40 studies on paediatric HIV infection, confirmed that youth had suboptimal medication adherence.

Adherence studies in Jamaican children, demonstrate better adherence in the residential care setting compared with home settings, due to directly observed therapy (DOT). Children are dependent on caregivers for the administration of doses, so adherence issues can be linked to caregiver related barriers in home settings. In addition, in adolescents, psychosocial factors correlated with poor adherence and virologic suppression. Psychosocial interventions including family and peer group therapy, improved adherence and virologic suppression.

A comprehensive assessment and preparation of the family for adherence should include the following:

- ✓ A multidisciplinary team approach including nursing, social, behavioural and psychological assessment.
- ✓ Identification of behavioural and social barriers to adherence and formulation of a plan to maximize adherence.
- ✓ Additional psychosocial intervention for non-adherent HIV infected mothers. Barriers to their care, directly impact her HIV infected infant/child.
- ✓ There should be intensive family education and medication training including measurement of doses of liquid preparations and anticipation of side effects.
- ✓ Teach pill swallowing (see Chapter 5).
- ✓ Medication Refusal: Once started, a child cannot be allowed to refuse medication. The child should not be allowed any other activity until the medication is taken.

- ✓ Ensure palatability or challenges with size are addressed.

Discuss all phases of medication administration, including:

- ✓ Who will pick up the medication?
- ✓ Where will it be stored?
- ✓ Who is responsible for administering medication to the child or for direct observation of the teen?
- ✓ Determine a suitable time/s for administration
- ✓ Determine an alternative plan if there is a change of routine (e.g., having a few doses in the event of an early departure from home or a late arrival back home)

Follow up every two weeks then monthly until the viral load is undetectable. After that visits can be every 2-3 months if there are no adherence issues.

Monitoring adherence should include:

1. The use of an objective measure to monitor adherence such as:
 - Prescription refill records
 - Adherence scores (see Appendix 6)
 - Pill counts
 - Home visits for families with identified barriers
 - All members of the team should include an adherence component in their follow up visit
 - DOT
2. The guardians should be specifically asked about medication intolerance, dosing difficulties or inconvenience
3. Ask the caregiver to describe the regimen in detail. How much is measured or what pills are given. Difficulty in outlining the regime suggests non-adherence
4. After two weeks after initiation of ARVs, there should be a rapid drop in viral

load and then a slow decline that continues for months. Viral load should be <400 copies by 2-3 months in most cases and < 50 copies by 6 months.

5. Failure to respond expectedly should prompt evaluation of adherence.
6. Refer to National Retention and Recovery for missed appointments. Take immediate action, as children and adolescents are a flight risk and must be closely followed/monitored.

Table 21: Assessment of Medication Adherence

Routine Assessment of Medication Adherence	Description
Monitor Viral Load	When initiating or changing ARVs, viral load monitoring should be done every 6 months until suppression.
Assess Quantitative Self-Report of Missed Doses	Ask the patient and/or caregiver about the number of missed doses over a defined period (1, 3, or 7 days).
Elicit a Description of the Medication Regimen	Ask the patient and/or caregiver about the name, appearance, and number of medications, and how often the medications are taken.
Assess Barriers to Medication Administration	Engage the patient and caregiver in a dialogue about facilitators and challenges to adherence.
Monitor Pharmacy Refills	Approaches include a pharmacy based or clinic based assessment of on time medication refills.

Conduct Announced, and Unannounced Pill Counts	Approaches include asking patients to bring medications to the clinic, home visits, or referral to community health aide.
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Table 22: Intervention Strategies to Improve Adherence

Intervention Strategies	Description
Implement DOT	Include brief hospitalization if indicated.
Use a Multidisciplinary Team Approach	Have more than one member of the multidisciplinary team monitor adherence at each visit and in-between visits by telephone, email, text, and social media, as needed.
Managed Problem Solving (MAPS)	A five-step process which consists of 1) identifying barriers to adherence, 2) brainstorming to generate potential solutions, 3) decision-making and developing a plan of action, 4) implementing the plan, and 5) evaluating and modifying the plan as necessary.
Peer Support	Provide access to support groups, peer groups, or one-on-one counselling for caregivers and patients, especially for those with known depression or psychosocial issues.
Patient Aides	Encourage use of pill boxes, calendars, mobile apps, alarms, and timers.

Provide Ongoing Support	Reward effort (hug the children and give them a treat!). Provide encouragement and understanding of the difficulties associated with maintaining adherence to daily medication regimens.
Consult a Psychologist and Social Worker	Incorporate behavioural interventions to improve adherence and address mental health or social barriers to care.

If non-adherence persists after intervention, refer all cases to the Child Protection and Family Services Agency (CPFSA).

GUIDELINE: ADHERENCE		
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CHAPTER 11: PEER SUPPORT

PEER SUPPORT

Peer support groups are safe spaces to share experiences. In addition to the other psychosocial support strategies, peer support can help ALHIV counter stigma and discrimination, cope with fear and hopelessness after diagnosis, improve adherence to care and treatment services, and deal with issues like disclosure to partners, friends, and family.

Peer Educators are valuable members of the treatment care and support network as they expand the ability to provide quality care to adolescents and caregivers. Where health facility staff are pressed for time to conduct educational activities, supportive peers and support groups can invest more time to share information and personal experiences.

These sessions can occur during the clinic, as outreach activities, or as a separate activity outside of clinic hours. Peer Navigators are available at most sites to assist these adolescence and caregivers with accessing these treatment care and support services. Please see reference for MOHW guidelines for support groups.

GUIDELINE: PEER SUPPORT		
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CHAPTER 12: CARE OF THE FAMILY AFFECTED BY HIV DISEASE

Care of the Family Affected by HIV Disease

- Family-focused care means that all members of the multidisciplinary care team think about the needs of all family members and not just those of the adolescent client.
- It also means thinking about the linkages between the individual client, the client's family, and the community. Identifying resources available in the community and exploring how these services can meet the needs of families.
- Health care workers can work with families or peer educators to make a list of community resources. The resource directory could include, nutritional and food support, spiritual guidance and support, transportation to get to the clinic, skills training, job placement, income generating activities, social grants, education, among others.
- Depending on the client's age and family situation, health workers should make it a routine practice to ask the adolescent about caregivers and other family members. They should also encourage the client to bring family members to the clinic for services if needed. Health workers can provide family members with ongoing education and information on HIV care and treatment, adherence counselling and support, and general support on caring for ALHIV.
- With older adolescents, health workers should also enquire about partners and children. When the adolescent is ready, they should be encouraged and supported to bring their partner(s) to the clinic for information on HIV, safer sex, and HIV testing and treatment.
- It is important to recognize the burnout of caregivers. Make appropriate referrals to care for the parent or guardian who may have psychosocial issues that could have an indirect impact on adherence in the child.

GUIDELINE: CARE OF THE FAMILY WITH HIV DISEASE		
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CHAPTER 13: DISPLACED HIV INFECTED CHILDREN

Displaced HIV Infected Children

HIV infected children or adolescents, may be separated from their parents for a variety of reasons; including death, abandonment, migration, a change in adult relationships, or financial difficulties.

The best place for an orphaned or separated child is within their extended families. Grandmothers seldom turn children away, and children may be fostered by non-relatives who can provide a better quality of life than their relatives.

A goal for a shift in caregiving is to improve the standard of living for the child, improve school attendance, and overall health. Placement in an institutionalized setting should be for a short term while suitable family placement is being arranged.

With appropriate external support (see also family focused care), families can cope with and successfully care for the HIV infected child and adolescent.

Once it is established who the new primary caregiver is, there should be full disclosure of the child's status and education of the caregiver to facilitate comprehensive treatment and viral suppression. Discuss with whom disclosure is safe to maintain within the child's or adolescent's new home environment.

Within the institutionalized setting, there should be disclosure and confidentiality among caregivers who are trained to administer ARVs in a supportive environment. The importance of confidentiality should be stressed as children/adolescents may be the subject of bullying and discrimination by other wards of the state.

GUIDELINE: DISPLACED HIV INFECTED CHILDREN		
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CHAPTER 14: APPENDICES

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- Appendix 10: Opportunistic Infections
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- Appendix 13: Social Support for Vulnerable Children
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Appendices

Appendix 1: Breastfeeding in Special Circumstances

What to Do if a Mom Chooses to Breastfeed

The national guidelines, recommend formula feeding (breast milk replacement for HIV exposed infants).

If a mother chooses to breastfeed, she should receive intensive counselling, which is patient-focused and evidenced-based. If she still chooses to breastfeed, this **MUST** be done under specific circumstances. She must be fully adherent on ARVs with an undetectable viral load.

Exclusive breastfeeding means that the infant receives only breast milk. No other liquids or solids are given, not even water, except oral rehydration solution, or drops/syrups of vitamins, minerals or medicines.

Breastfeeding is exclusive for up to 6 months postpartum, followed by the introduction of complementary foods.

The exclusively breastfed HIV exposed infants should be maintained on newborn prophylaxis and have PCR testing every 3 months throughout breastfeeding and 6 weeks, 3 months, and 6 months after breast milk cessation **PLUS** a rapid test at 18 to 24 months of age.

Maternal viral load should be monitored more frequently up to every 2-3 months.

Breast milk replacement is commenced only immediately after cessation of breast feeding.

Treat maternal mastitis and infant thrush promptly.

Infant Feeding and Transmission of Human Immunodeficiency Virus in the United States Committee of Pediatric Aids. Pediatrics Feb 2013, 131 (2) 391-396; DOI: 10.1542/peds.2012-3543

Appendix 2: CDC Clinical category

Category	Clinical Manifestations
Mildly Symptomatic (A)	Children with two or more of the conditions listed, but none of the conditions listed in the Moderate Symptoms category: • Lymphadenopathy (lymph nodes are ≥ 0.5 cm at more than two sites and/or bilateral at one site) • Hepatomegaly • Splenomegaly • Dermatitis • Parotitis • Recurrent or persistent upper respiratory tract infection, sinusitis, or otitis media
Moderately Symptomatic (B)	• Anemia (hemoglobin < 8g/dl in children aged 6 months) • Cardiomyopathy • Cytomegalovirus infection (before age 1 month) • Diarrhoea, recurrent or chronic • Hepatitis • HSV stomatitis, recurrent (more than two episodes within 1 year) • HSV bronchitis, pneumonitis, or esophagitis with onset before age 1 month • Herpes zoster (shingles) involving at least two distinct episodes or more than one dermatome • Leiomyosarcoma • Oral Candidiasis

Category	Clinical Manifestations
	• Lymphoid interstitial pneumonia or pulmonary lymphoid hyperplasia complex • Nephropathy • Nocardiosis • Persistent fever (lasting >1 month) • Toxoplasmosis, onset before age 1 month • Varicella, disseminated (complicated chicken-pox)
AIDS-Defining Conditions (C)	• Cryptosporidiosis, chronic intestinal (>1-month duration) • CMV disease (other than the liver, spleen, or lymph nodes), onset at age >1 month • CMV retinitis (with loss of vision) • Encephalopathy attributed to HIV * • HSV: chronic ulcers (>1-month duration) or bronchitis, pneumonitis, or esophagitis (onset at age >1 month) • Histoplasmosis, disseminated or extrapulmonary • Isosporiasis, chronic intestinal (>1-month duration) • Kaposi sarcoma • Lymphoma (Burkitt, immunoblastic, brain) • Extrapulmonary Cryptococcosis • Mycobacterium avium complex or Mycobacterium kansasii, disseminated or extrapulmonary

Category	Clinical Manifestations
	• Mycobacterium tuberculosis of any site, pulmonary, disseminated, or extrapulmonary • Mycobacterium, other species or unidentified species, disseminated or extrapulmonary • Pneumocystis jirovecii (previously known as Pneumocystis carinii) pneumonia • Pneumonia, recurrent (≥ 6 years) • Progressive multifocal leukoencephalopathy • Salmonella septicemia, recurrent • Toxoplasmosis of the brain, onset at age >1 month • Wasting syndrome attributed to HIV**

*Encephalopathy (at least one of the following progressive findings present for at least 2 months in the absence of a concurrent illness other than HIV infection that could explain the findings): a failure to attain or loss of developmental milestones or loss of intellectual ability, verified by standard developmental scale or neuropsychological tests; b impaired brain growth or acquired microcephaly demonstrated by head circumference measurements or brain atrophy demonstrated by computerized tomography or magnetic resonance imaging (serial imaging is required for children under 2 years of age; c acquired symmetric motor deficit manifested by two or more of the following: paresis, pathologic reflexes, ataxia, or gait disturbance

** Wasting syndrome in the absence of a concurrent illness other than HIV infection that could explain the following findings: a) persistent weight loss more than 10% of baseline b) downward crossing of at least two of the following percentile lines on the weight-for-age chart (e.g., 95th, 75th, 50th, 25th, 5th) in a child at least 1 year of age or c) less than the 5th percentile on weight-for-height chart on two consecutive measurements at least 30 days apart PLUS a) chronic diarrhoea (i.e., at least two loose stools per day for more than 30 days) or b) documented fever (for at least 30 days, intermittent or constant)

Appendix 3: CDC Immune Category

Age on date of CD4 test

Immune Category	≤ 12 months cell/μL (%)	1-<6 years cell/μL (%)	≥ 6 years cell/μL (%)
Stage 1	≥1500(≥ 34)	≥1000(≥ 30)	≥500(≥ 26)
Stage 2	750-1499 (26-33)	500-999(22-29)	200-499(14-25)
Stage 3	< 750 (<26)	<500 (<22)	<200(<14)

N.B. The stage is based primarily on the CD4+ T-lymphocyte count; the CD4+ T-lymphocyte count takes precedence over the CD4 T-lymphocyte percentage, and the percentage is considered only if the count is missing. Once an HIV infection has been staged into a higher clinical category, it remains in that category permanently. For example, a person diagnosed at stage 3, will remain classified as stage 3 even with improvements in CD4 counts. In addition, the infection is classified based on the lowest CD4⁺ T-cell count in that patient.

Appendix 4: HEADSS Assessment Tool

Category	Description
Home	Ask about relationships with each member of the household, i.e., how do you get along with...; do you think your parent understands you, cares about you? Listens to what you have to say?
Education	School name, grade, has repeated a grade; what are their grades on average; have they ever been suspended or expelled? How do they get along with teachers? Classmates?
Eating	What is their intake like? How many meals? Snacks? Are there food security concerns? i.e., Accessing food. How do they feel about their bodies? Are they doing anything to change their weight? If yes, then what?
Activities	Do they have any close friends that they trust? What do they do when they hang out with friends? Are they involved in extracurricular activities?
Drugs	Do their friends use? Personal use?
Sexuality and Sex	Orientation, 1 st intercourse, # of partners, type of partners, condom use, risk behaviour, etc.
Safety	Abuse concerns
Suicide/Mood	How do they usually feel? Do they ever feel so down they feel like killing themselves

Appendix 5: Psychosocial Assessment Tool

Client Name: _____ Client File#: _____

Date: _____

Questions	Notes/Response
1. Smile, introduce yourself and give a short explanation of your role. Explain that this discussion will be confidential.	
2. Can you tell me how things have been going since you learned your HIV-status (or since we last met)? How are you coping? Explore and discuss client's coping strategies	
3. Tell me about your mood now. Do you feel sad or stressed? What changes have you noticed in your mood? What about your eating and sleeping habits? Assess the risk of depression and need for referral	
4. How often in the last week, have you used cigarettes, alcohol, or other drugs? Assess for harmful coping strategies, such as drug/alcohol use, provide counselling and referrals	

Questions	Notes/Response
5. To whom have you disclosed your HIV-status? What was their reaction? Do you want to disclose it to anyone else? What concerns do you have about disclosure? <i>Counsel on disclosure</i>	
6. To whom do you feel close? Whom can you go to for emotional support? <i>Counsel on the importance of social support</i>	
7. Do you belong to a community/religious organization or support group? Would you be willing to join a support group to meet other ALHIV? <i>Make referrals as needed</i>	
8. Tell me about any negative attitudes or treatment you have experienced. Has anyone caused you harm (e.g., been violent, made unwanted sexual advances)? <i>ounsel and discuss support services; consider gender-based violence services, if appropriate</i>	
9. Some adolescents have sex with their partners. It is important for you and your partner to do this safely. Are you having sex? If so, what are you doing to prevent pregnancy and the spread of STIs and HIV? <i>Screen for sexual risk-taking and counsel on safer sex, dual protection, etc.; give condoms</i>	

Questions	Notes/Response
10. Let us talk about your living situation. Whom are you living with? How long have you lived with them? How well do you get along? <i>Assess the living situation</i> <i>If not living with your parents, ask: Where are your parents? When did this happen? How did this affect you?</i>	
11. Tell me what you do on most days. Do you, for example, go to school or work outside the home? Where do you go to school/work? How is this going for you? <i>Assess school/work situation</i>	
12. Do you have financial support from your family or partner, a regular source of income, or do you receive help, such as social grants or food? <i>Refer to the social worker and community-level support</i>	
13. Other than coming to this clinic, where else do you go for health services (for example, other clinics, traditional healers, etc.)?	
14. How do you/will you remember to come to this clinic for your appointments and refills? How do you/will you manage it with your school or work? Who can help you? <i>Counsel on adherence to care</i>	

Questions	Notes/Response
15. How do you/will you remember to take your medications every day? How do you/will you remember when to come back to the clinic? Who can help you? Counsel on adherence and briefly discuss: • WHO will give or manage your medications? • WHEN will you take them? • WHERE will you store them? • HOW will you remember to take them (review use of reminders, like calendars, pillboxes, etc.)	
16. What other questions or concerns do you want to discuss today? Would you like to bring someone else into our conversation — today or at another visit (e.g., family member, partner)?	
17. Summarize the session and review immediate plans and next steps, including the next clinic visit date.	

Adapted from: ICAP. (2010). Improving retention, adherence, and psychosocial support within PMTCT services: A toolkit for health workers.

New York, NY: Columbia University/ICAP.

Appendix 6: Adherence Questionnaire

Person completing questionnaire: _____

Date: _____

Adherence Questionnaire

Client Name: _____ Tel #: _____

Docket Number: _____

Most recent CD4 test date: _____

CD4 Results: _____

Most recent VL test date: _____

VL Results: _____

Current ART Regime: _____

Date Regime started: _____

Circle the number that corresponds to the client's answer. The number assigned to the answers is also allotted score. The total score is the sum of the individual scores.

1. How often do you feel that you have difficulty taking your HIV medication on time?

- [1] All of the time
- [2] Most of the time
- [3] Rarely
- [4] Never

2. On average, how many days PER WEEK would you say that you missed at least one dose of your HIV medications or did not take it as prescribed?

- [1] Every day
- [2] 4 to 6 days/week
- [3] 2 or 3 days/week
- [4] Once a week
- [5] Less than a week
- [6] Never

3. During the last 7 days, would you say you:

- [1] Took none of your pills
- [2] Missed more than one doses
- [3] Missed one dose
- [4] Took all of your pills every day

4. What were the reasons for non-adherence? (For any reason indicated, please refer client to Social Worker).

☐	Side effects
☐	Lack of food
☐	Fare/transportation (to collect medication from the pharmacy)
☐	Forgot
☐	Work
☐	Confidentiality (at home, work, etc.)
☐	Travel (left home for a few days)
☐	Mental Health (stressed out, depression, etc.)
☐	Sick
☐	Homeless
☐	Don't think it is necessary
☐	Religion (faith, cured, etc.)
☐	Proper medication adherence was not explained
☐	Taking herbs, other medication, etc
☐	Sharing of medication
☐	Stressed out
☐	Tired of taking pills
☐	Other: _____

Score: ____/14 (For scores 10 and below, please refer to the Social Worker.) Was pill-count done? Circle yes or no.

Appendix 7: ARV Dosing

Doses need to be adjusted for growth in children until the maximum dose is achieved

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT																						
I. Abacavir 60MG + Lamivudine 30MG Tablets/60PK Dispersible	Abacavir 8 mg/kg twice daily (maximum 300 mg per dose) or Abacavir 16 mg/kg once daily (maximum 600 mg per dose) (see Lamivudine below) <table><tr><th rowspan="2">Weight</th><th colspan="2">Twice daily</th><th rowspan="2">Once daily</th></tr><tr><th>AM</th><th>PM</th></tr><tr><td>14-<20kg</td><td>½(150)</td><td>½(150)</td><td>1 tablet</td></tr><tr><td>≥20-25kg</td><td>½(150)</td><td>1(300)</td><td>1½ tablets</td></tr><tr><td>≥25kg</td><td>1(300)</td><td>1(300)</td><td>2 tablets</td></tr><tr><td>≥25kg</td><td colspan="3">1 tablet once daily</td></tr></table>	Weight	Twice daily		Once daily	AM	PM	14-<20kg	½(150)	½(150)	1 tablet	≥20-25kg	½(150)	1(300)	1½ tablets	≥25kg	1(300)	1(300)	2 tablets	≥25kg	1 tablet once daily			Hypersensitivity reactions (HSRs) can be fatal. HSRs usually occur during the first few weeks of starting therapy. Symptoms may include fever, rash, nausea, vomiting, malaise or fatigue, loss of appetite, and respiratory symptoms (e.g., cough, shortness of breath). Permanently discontinue if HSR occurs
Weight	Twice daily		Once daily																					
	AM	PM																						
14-<20kg	½(150)	½(150)	1 tablet																					
≥20-25kg	½(150)	1(300)	1½ tablets																					
≥25kg	1(300)	1(300)	2 tablets																					
≥25kg	1 tablet once daily																							
II. Abacavir Sulphate Tablet 300MG/60Pk		<i>Do not use abacavir /lamivudine in patients with impaired hepatic function as abacavir dose cannot be adjusted.</i>																						

ARV FORMULATIONS	PAEDIATRIC DOSING		COMMENT
III. Abacavir Sulphate/ Lamivudine Tablet 600/ 300MG/30Pk	Dose adjustments can be made using dispersables for patients with a creatine clearance <50 ml/min or patients on dialysis.		<i>Do not use</i> abacavir /lamivudine in patients with a creatinine clearance <50 ml/min or patients on dialysis as lamivudine dose cannot be adjusted.
IV. Atazanavir 300mg/ Ritonavir 100mg/ 30pk	Weight	Once Daily	• Indirect hyperbilirubinemia • Prolonged PR interval, first-degree symptomatic AV block in some patients • Nephrolithiasis • Increased serum transaminases • Hyperlipidaemia Atazanavir absorption is dependent on low gastric pH; adjust the dose if given with a drug that alters the pH. Use with caution in patients with mild or moderate hepatic impairment and do not use in severe hepatic impairment.
	< 15 kg	Not recommended	
	15-<35kg	Atazanavir/ritonavir 200 mg/100 mg, once daily with food	
	≥35kg	Atazanavir/ritonavir 300 mg/100 mg, both once daily with food	
	≥25kg	1 tablet once daily	

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT
		Because atazanavir can prolong the PR interval, use atazanavir with caution in patients with pre-existing cardiac conductions, systemic disease or with other drugs that are known to pro-long the PR interval (e.g., calcium channel blockers, be-ta-blockers, digoxin, verapamil).
Darunavir Tablet 600MG/60 PK	Adolescent (Aged ≥ 12 years and weighing ≥ 40 kg) Darunavir 600 mg plus ritonavir 100 mg, both twice daily with food	Darunavir must be administered with food Use cautiously in patients with hepatic impairment. Not recommended in patients with severe hepatic impairment. No dose adjustment with moderate renal impairment (creatinine clearance 30–60 mL/min) <i>Do not use</i> darunavir in children aged < 3 years or weighing ≤ 10kg

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT										
I. Dolutegravir 50mg (30 tabs/bottle)	Children ≥ 20 kg Dolutegravir 50 mg once daily (current WHO recommendation)	Insomnia Headache Neuropsychiatric symptoms (i.e., depression and/or suicidal thoughts or actions), especially in patients with a history of psychiatric illness.										
II. Dolutegravir 10 mg, 25 mg	Canadian and European approved dosing <table><tr><th>Bodyweight (kg)</th><th>Once daily dose regime</th></tr><tr><td>15 to < 20</td><td>20 mg (two 10 mg tablets)</td></tr><tr><td>20 to < 30</td><td>25 mg</td></tr><tr><td>30 to < 40</td><td>35 mg (one 25 mg and one 10 mg tablet)</td></tr><tr><td>≥40</td><td>50 mg (one 50 mg tablet)</td></tr></table> Increase the weight-based dose to twice daily if DTG is co-administered with Efavirenz, Rifampin, Fosamprenavir/ritonavir, Tipranavir/ritonavir	Bodyweight (kg)	Once daily dose regime	15 to < 20	20 mg (two 10 mg tablets)	20 to < 30	25 mg	30 to < 40	35 mg (one 25 mg and one 10 mg tablet)	≥40	50 mg (one 50 mg tablet)	Rare cases of hypersensitivity reactions, including rash and drug reaction (or rash) with eosinophilia and systemic symptoms, constitutional symptoms, and organ dysfunction Do not use in patients with severe hepatic or renal impairment. No dose adjustment required in mild or moderate hepatic or renal impairment
Bodyweight (kg)	Once daily dose regime											
15 to < 20	20 mg (two 10 mg tablets)											
20 to < 30	25 mg											
30 to < 40	35 mg (one 25 mg and one 10 mg tablet)											
≥40	50 mg (one 50 mg tablet)											

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT															
Efavirenz Tablet 600MG/30PK	<i>Not recommended in children < 3 years</i> <i>Child and Adolescent (Weighing ≥ 40 kg)</i> <i>Dose: Efavirenz 600mg once daily</i>	Central nervous system (CNS) symptoms, such as fatigue, poor sleeping patterns, insomnia, vivid dreams, impaired concentration, agitation, seizures, depression, suicidal ideation Hepatotoxicity; gynecomastia Bedtime dosing is recommended, particularly during the first 2 to 4 weeks of therapy, to improve the tolerability of CNS side effects															
Etravirine 100MG/ Tablet/120 PK	<table border="1"> <thead> <tr> <th>Bodyweight (kg)</th><th>AM (mg)</th><th>PM (mg)</th></tr> </thead> <tbody> <tr> <td>10 to < 20</td><td>100</td><td>100</td></tr> <tr> <td>20 to < 25</td><td>125</td><td>125</td></tr> <tr> <td>25 to < 30</td><td>150</td><td>150</td></tr> <tr> <td>≥ 30</td><td>200</td><td>200</td></tr> </tbody> </table>	Bodyweight (kg)	AM (mg)	PM (mg)	10 to < 20	100	100	20 to < 25	125	125	25 to < 30	150	150	≥ 30	200	200	Always administer etravirine with food. Adverse events include: • Nausea • Diarrhoea • Rash, including Stevens-Johnson syndrome
Bodyweight (kg)	AM (mg)	PM (mg)															
10 to < 20	100	100															
20 to < 25	125	125															
25 to < 30	150	150															
≥ 30	200	200															

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT												
Etravirine 100MG/ Tablet/120 PK (Cont'd)		Hypersensitivity with rash, constitutional findings, and sometimes organ dysfunction, including hepatic failure. No dose adjustment in patients with hepatic or renal impairment.												
I. Zidovudine 60MG + Lamivudine 30MG + Nevirapine 50MG/60 PK Dispersible II. Zidovudine 300MG + Lamivudine 150MG + Nevirapine 200MG/60 PK III. Zidovudine 60MG Lamivudine 30MG + /60 PK Dispersible	See dosing of zidovudine, lamivudine, nevirapine.	Approximate individual drugs closely to fixed dose combination.												
I. Zidovudine Oral Solution 50MG/5ML/240mls	<table border="1"> <thead> <tr> <th>Weight</th><th>Twice daily AM PM</th><th>Once daily</th></tr> </thead> <tbody> <tr> <td>14-<20kg</td><td>½ (75) ½ (75)</td><td>1 tablet</td></tr> <tr> <td>≥20-25kg</td><td>½ (75) 1(150)</td><td>1 ½ tablet</td></tr> <tr> <td>≥25kg</td><td>1(150) 1(150)</td><td>2 tablets</td></tr> </tbody> </table>	Weight	Twice daily AM PM	Once daily	14-<20kg	½ (75) ½ (75)	1 tablet	≥20-25kg	½ (75) 1(150)	1 ½ tablet	≥25kg	1(150) 1(150)	2 tablets	Dose adjustment is required in patients with renal insufficiency.
Weight	Twice daily AM PM	Once daily												
14-<20kg	½ (75) ½ (75)	1 tablet												
≥20-25kg	½ (75) 1(150)	1 ½ tablet												
≥25kg	1(150) 1(150)	2 tablets												

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT
II. Lamivudine 150MG/Tablet/60 PK	Aged ≥ 4 Weeks to < 3 Months: Lamivudine 4 mg/kg/dose twice daily of the oral solution. Aged 3 Months to < 3 Years: Lamivudine 5 mg/kg twice daily of the oral solution (maximum of 150 mg per dose) Aged ≥ 3 Years: Lamivudine 5 mg/kg twice daily of the oral solution (maximum 150 mg per dose) or 10 mg/kg once daily of the oral solution (maximum 300 mg per dose)	Severe exacerbation of hepatitis can occur in patients with hepatitis B virus (HBV) and HIV coinfection who discontinue lamivudine
Lopinavir 100MG/ Ritonavir 25MG Tablet/60 PK Lopinavir 200MG/ Ritonavir 50MG Tablet/120PK	LPV/r 300 mg/75 mg per meter squared of body surface area per dose twice daily. (Aged 14 Days–12 Months) This approximates LPV/r 16 mg/4 mg (both per kg body weight) twice daily.	Younger children may be started on LPV/r oral solution but transitioned to the more convenient LPV/r 100 mg/25 mg tablets when they weigh 10 kg or more and can swallow tablets

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT
II. Nevirapine Tablet 200MG/60 PK (Cont'd)	(maximum dose of 200 mg twice daily). When adjusting the dose for a growing child, the mg dose needs not be decreased as the child reaches age 8 years; Rather, the mg dose can be left static to achieve the appropriate mg/m² dose as the child grows, as long as there are no adverse effects.	with potential for multisystem organ involvement and shock. Nevirapine associated skin rash usually occurs within the first 6 weeks of therapy. If a rash occurs during the initial 14-day lead-in period, do not increase the dose until the rash resolves. Most cases of Nevirapine associated hepatic toxicity occurs during the first 12 weeks of therapy; clinical and laboratory monitoring, including liver function tests, is important during this period.

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT
II. Nevirapine Tablet 200MG/60 PK (Cont'd)		Do not initiate in (ARV)-naive Tanner 4 adolescent females with pre-Nevirapine CD4 T lymphocyte (CD4) cell, counts >250 cells/mm ³ and in ARV-naive Tanner 4 adolescent males with pre-Nevirapine CD4 counts >400cells/mm ³
Raltegravir Tablet 400MG/60 PK	Adolescent > 50 kg 400 mg twice daily Suspension for neonates(>2kg) and chewable tablets for children at 6 mg /kg /dose twice daily	No dose adjustment in patients with renal impairment or mild to moderate hepatic insufficiency. Avoid in patients with hepatic impairment.
I. Tenofovir 300M+ Lamivudine 300MG/ 30PK II. Tenofovir 300MG/ Emtricitabine 200MG/ 30 PK	Tenofovir 8 mg/kg once daily Lamivudine 8 mg /kg once daily Child and Adolescent (Weighing ≥35 kg) One 300 mg/200 mg tablet twice daily.	For adolescents under 35 kg, alternating one tablet one day with the half tablet the next day can be used to prevent toxicity. Take with or without food

ARV FORMULATIONS	PAEDIATRIC DOSING	COMMENT								
Tenofovir 300MG + Lamivudine 300MG + Efavirenz 600MG/30PK	Child and Adolescent (Weighing ≥40 kg) One tablet once daily	Take on an empty stomach								
Tenofovir 300mg/ Lamivudine 300mg/ Dolutegravir 50mg/30pk	Children weighing >20kg and >9 years old One tablet once daily	See individual drug requirements								
I. Zidovudine Oral Solution 50MG/5ML/240mls II. Zidovudine 300MG Tablet/60 PK	<table><tr><th>Weight</th><th>Twice daily dosing</th></tr><tr><td>4- < 9kg</td><td>12 mg/kg/dose</td></tr><tr><td>9- <30kg</td><td>9mg/kg/dose</td></tr><tr><td>≥ 25kg</td><td>300mg</td></tr></table> OR 180-240 mg/m²/dose twice daily	Weight	Twice daily dosing	4- < 9kg	12 mg/kg/dose	9- <30kg	9mg/kg/dose	≥ 25kg	300mg	Dose adjustments are required in a patient with renal impairment and may be needed for hepatic impairment
Weight	Twice daily dosing									
4- < 9kg	12 mg/kg/dose									
9- <30kg	9mg/kg/dose									
≥ 25kg	300mg									
Zidovudine Tablet 300mg + Lamivudine 150 MG/60PK	Child and Adolescent (Weighing ≥30 kg) One 300mg/150 mg tablet twice daily	Do not use in patients with creatinine clearance <50 mL/min or in patients who are on dialysis or who have impaired hepatic function.								

See Chapter on HIV diagnosis for tables on Medication Toxicity and Management

Appendix 8: Laboratory Results

Interpretation of Abnormal Lab Results

Lab test	Interpretation of Abnormalities
CBC	Zidovudine - associated anaemia, leukopenia, and macrocytosis Nutrition associated micronutrient deficiency (do further evaluation and include iron and folate if nutritional aetiology suggested)
Electrolytes with ↑ Anion Gaps	Nucleoside reverse transcriptase inhibitor-associated lactic acidosis AZT; less likely with 3TC, FTC, ABC, TAF, and TDF; secondary to mitochondrial dysfunction
↓Ca PO₄ ↑Alk Phos ↑Cr	Hypophosphatemia is an early feature of tubular dysfunction and associated bone demineralization or nephropathy in clients on TDF; this occurs before changes in serum Ca, alkaline phosphatase, electrolyte and protein wasting, and elevation of creatinine or impaired creatinine clearance DTG, COBI, RPV can cause ↑Creatinine
↑ Glucose	Due to protease inhibitors (LPV/r or ATZ/r). AZT. Do additional profile for diabetes
↑Transaminases	Many ARV drugs, e.g., NVP, EFV, ABC, RAL, DTG, and MVC associated with HSR. Strong association especially with NVP and EFV Patients with HBV coinfection may develop flare of hepatitis with the initiation or withdrawal of 3TC, FTC, TDF, or TAF. Flare can occur with the emergence of 3TC resistance Hepatitis may be a manifestation of IRIS if it occurs early in therapy, especially in patients with HBV or HCV coinfection

Lab test	Interpretation of Abnormalities
↑ Bilirubin	Atazanavir
Glycosuria, Proteinuria, ↑ Urinary Protein/ Creatinine Ratio	Early indicator of renal dysfunction due to TDF.
↓ Serum Albumin and/or Proteins	May be related to GI loss with chronic diarrhoea, renal loss of protein with HIV nephropathy, or nutrition-related. Evaluate and treat the specific cause. If nutrition-related, refer to a dietician for an increased protein diet
Abnormal Lipid Profile	Associated with NRTI (zidovudine and lamivudine) but more prominent with protease inhibitors. Uncontrolled HIV disease is also associated with dyslipidaemia.
↑ Amylase and Lipase	Rarely with RTV-boosted PIs or NRTIs

Discordant Virologic, Immunologic and Clinical Response in HIV Positive Children and Adolescents	Causes of Abnormalities
Poor Immunologic Response Virologic Suppression and Good Clinical Response:	• Lab error (in CD4 or viral load measurement) • Misinterpretation of normal, age-related CD4 decline • Low pre-treatment CD4 cell count or percentage • Adverse effects of using AZT • Use of systemic corticosteroids or chemotherapeutic agents • Conditions that can cause low CD4 values, such as HCV, acute viral infections, TB, malnutrition, Sjogren's syndrome, sarcoidosis, and syphilis
Poor Immunologic Response Poor Clinical Response Virologic Suppression:	• Lab error • Falsely low viral load result for an HIV strain/type that is not detected by viral load assay (HIV-1 non-M groups, non-B subtypes; HIV-2) • Persistent immunodeficiency soon after initiation of ARV but before ARV-related reconstitution • Primary protein-calorie malnutrition • Untreated TB • Malignancy

Discordant Virologic, Immunologic and Clinical Response in HIV Positive Children and Adolescents	Causes of Abnormalities
Poor Clinical Response Adequate Virologic Adequate Immunologic Responses	• IRIS • A previously unrecognized, pre-existing infection or condition (e.g., TB, malignancy) • Malnutrition • Clinical manifestations of previous organ damage: brain (e.g., strokes, vasculopathy), lungs (e.g., bronchiectasis), cardiac(cardiomyopathy), renal (HIV-related kidney disease) • A new clinical event due to a non-HIV illness or condition

Management of Abnormal ARV Related Lab Results

Adverse Event	Presentation	Management
Anemia	Asymptomatic; mild fatigue; Pallor; tachypnoea Rarely CCF	In neonates, anaemia due to AZT requires intervention if haemoglobin <7.0 g/dL. AZT administration can be limited to 4 weeks in low-risk neonates. In children discontinue non ARV, marrow toxic drugs, if feasible. Treat coexisting iron deficiency, OIs, and malignancies. For persistent, severe anaemia that is thought to be associated with ARV drugs (typically macrocytic anaemia), switch to a regimen that does not contain AZT.
Macrocytosis	Asymptomatic; mean cell volume >100fl	No management required.
Hepatitis	Asymptomatic elevation of AST and ALT Symptomatic hepatitis with nausea, fatigue, and jaundice May present in the context of HSR with rash, lactic acidosis, and Hepatic steatosis.	Evaluate the patient for other infectious and non-infectious causes of hepatitis and monitor the patient closely. Asymptomatic Hepatitis: Potentially offending ARV drugs should be discontinued if ALT or AST level is >5 times ULN. Symptomatic Hepatitis: Discontinue all ARV drugs and other potentially hepatotoxic drugs. If a patient experiences hepatitis that is attributed to NVP, NVP should be permanently discontinued. Consider viral causes of hepatitis HAV, HBV, HCV, EBV, and CMV.

Adverse Event	Presentation	Management
Indirect Hyperbilirubinemia	May be asymptomatic or associated with jaundice Levels of direct bilirubin may be normal or slightly elevated when levels of indirect bilirubin are very high. Normal AST and ALT	Isolated indirect hyperbilirubinemia is not an indication for the cessation of the potentially offending ARV drug. Jaundice may result in non-adherence, particularly in adolescents, this side effect should be discussed.
Insulin Resistance, Asymptomatic Hyperglycaemia, DM	Asymptomatic fasting hyperglycaemia (which sometimes occurs in the setting of lipodystrophy), metabolic syndrome, or growth delay Symptomatic DM (rare)	Counsel patient on lifestyle modification. Recommend that the patient consult with a dietician. If the patient is receiving AZT change to TAF, TDF, or ABC. For either RBG ≥ 200 mg/dL plus Symptoms of DM or FPG ≥ 126 mg/dL: the patient meets diagnostic criteria for DM. Consult Paediatrician/endocrinologist/internist. FPG 100–125 mg/dL: Impaired FPG suggests insulin resistance consult Paediatrician/endocrinologist/internist. FPG < 100 mg/dL: This FPG is normal, but a normal FPG does not exclude insulin resistance. Recheck FPG in 6 months–12 months.

Adverse Event	Presentation	Management
Lactic Acidosis	Asymptomatic or Insidious Onset: Generalized fatigue, weakness, and myalgias; vague abdominal pain, weight loss, unexplained nausea or vomiting; dyspnoea; peripheral neuropathy Patients may present with acute multi-organ failure	Symptomatic patients should include serum bicarbonate, anion gap, and/or arterial blood gas; amylase and lipase; serum albumin; hepatic transaminases and serum lactate. Temporarily discontinue all ARVs and provide supportive care (IV fluids, O₂) Following the resolution of clinical and laboratory abnormalities, resume therapy, either with an NRTI-sparing regimen or a revised NRTI-containing regimen. Institute a revised NRTI containing regimen with caution, using NRTIs that are less likely to induce mitochondrial dysfunction (ABC, TAF, or TDF preferred; possibly FTC or 3TC). Lactate should be monitored monthly for ≥3 months
Renal Dysfunction	More Common: Increased serum creatinine, proteinuria, normoglycaemic glucosuria Increased urinary protein/creatinine ratio Hypophosphatemia, usually asymptomatic; may present with bone and muscle pain, or muscle weakness.	Increased risk in children with the following: Aged >6 years; black race; advanced HIV; hypertension; diabetes; concurrent use of PIs (especially LPV/r) and pre-existing renal dysfunction. Risk increases with a longer duration of TDF treatment.

Adverse Event	Presentation	Management
	Less Common: Renal failure, acute tubular necrosis, Fanconi syndrome, proximal renal tubulopathy, interstitial nephritis, nephrogenic diabetes.	If TDF is the likely cause, consider using an alternative ARV drug. TAF has significantly less toxicity than TDF. Consult a nephrologist.
Elevation in Serum Creatinine	Asymptomatic. The drugs (DTG, COBI, RPV) decrease renal tubular secretion of creatinine, leading to an increase in serum creatinine levels without a true change in eGFR. A true change in the GFR may be associated with other medical conditions, the continuing rise of serum creatinine levels over time, and albuminuria.	No need to change therapy. Reassure the patient about the benign nature of the laboratory abnormality. Refer all other patients to the nephrologist if other medical condition suspected

Adverse Event	Presentation	Management
Dyslipidaemia	PIs: ↑ LDL-C, TC, and TG NNRTIs: ↑ LDL-C, TC, and HDL-C NRTIs: ↑ LDL- TC, and TG Occur in 10% to 20% in young children receiving LPV/r and in 40% to 75% of older children and adolescents with prolonged ARV history will have lipid abnormalities.	Refer patients if LDL-C ≥ 250 mg/dL or TG ≥ 500 mg/dL If LDL-C is ≥ 130 mg/dL and < 250 mg, or TG is ≥ 150 mg/dL and < 500 mg/dL implement diet, nutrition, and lifestyle management for 6 to 9 months. Consult with a dietician. If a 6-month to 9-month trial of lifestyle modification fails and the patient is aged ≥ 10 years, consider lipid-lowering therapy after consulting a specialist. The long-term risks of lipid abnormalities in children receiving ARV are unclear. However, persistent dyslipidaemia in children may lead to premature CVD. Consider using DTG or a PI with a lower prevalence of dyslipidaemia (ATV or DRV).
Pancreatitis	Emesis, abdominal pain, elevated amylase and lipase levels Asymptomatic hyperamylasaemia or elevated lipase does not in and of themselves indicate pancreatitis	Discontinue offending agents and avoid reintroduction. Manage symptoms of acute episodes. If pancreatitis is associated with hypertriglyceridemia, consider using interventions to lower TG levels.

Appendix 9 – ARV Reactions

Clinical Adverse Event and Presentation	Signs and Symptoms	Associated ARV	Drug Management
Global CNS Depression	Neonates/Premature Infants: Global CNS depression especially in LBW, premature or < 14 days.	LPV/r oral solution	Avoid the use of LPV/r until a postnatal age of 42 weeks and a postnatal age of ≥ 14 days. Discontinue LPV/r, and symptoms should resolve in 1–5 days. May be reintroduced outside the vulnerable window
Neuropsychiatric and CNS manifestations (CNS symptoms were uncommonly reported in early clinical experience in children and adolescents.)	Neuropsychiatric Symptoms: depression or exacerbation of preexisting depression; Anxiety; Suicidal ideation or attempted/completed suicide Other mild CNS symptoms Insomnia Dizziness Headache	DTG	For persistent or severe neuropsychiatric symptoms, consider discontinuation of DTG if a suitable alternative exists. For mild symptoms, continue DTG and counsel patient that symptoms will likely resolve with time.

Clinical Adverse Event and Presentation	Signs and Symptoms	Associated ARV	Drug Management
Nausea /Vomiting	Nausea, emesis, an-orexia and abdominal discomfort	All ARV drugs, but most notably RTV boosted PIs	Reassure patients that these adverse effects generally improve over time (usually 6–8 weeks). Provide supportive care In extreme or persistent cases, use antiemetics or switch to another ARV regimen.
Diarrhoea	More frequent soft stools (>3 per day)	All ARV drugs, but most notably RTV boosted PIs	If prolonged or severe, exclude infectious or noninfectious (e.g., lactose intolerance) causes of diarrhea. Reassure patient that this adverse effect generally improves over time (usually 6–8 weeks).

Clinical Adverse event and presentation	Signs and Symptoms	Associated ARV	Drug Management
Diarrhoea		All ARVs	Consider switching to another ARV regimen in persistent and severe cases If ARV cannot be changed using dietary modification; bulk-forming agents (e.g., psyllium) or antimotility agents (e.g., loperamide)
Central Lipohypertrophy	Central fat accumulation with increased abdominal girth, which may include a dorsocervical fat pad (buffalo hump). Gynecomastia in males or breast hypertrophy in females, particularly with the use of EFV.	PI or EFV. May occur in the absence of ARVs	Counsel patient on lifestyle modification and dietary interventions Stop smoking if relevant Consider using an INSTI (DTG) instead of a PI or EFV.

Clinical Adverse event and presentation	Signs and Symptoms	Associated ARV	Drug Management
Facial/Peripheral Lipoatrophy	Thinning of subcutaneous fat in the face, buttocks, and extremities, Preservation of lean body mass distinguishes lipoatrophy from HIV associated wasting	AZT	Replace AZT with another NRTI if possible.
Urolithiasis/ Nephrolithiasis	Crystalluria; Hematuria; Pyuria; Flank pain; Increased creatinine in some cases	ATV DRV causes crystalluria but not nephrolithiasis	Provide adequate hydration and pain control. Consider using another ARV in place of ATV.
Osteopenia and Osteoporosis	Usually asymptomatic Rarely presents as osteoporosis, a clinical diagnosis defined by evidence of bone fragility (e.g., fracture with minimal trauma)	Any ARV. Specifically, TDF when used in a regimen with RTV and PIs especially LPV/r	Ensure that the patient has sufficient intake and levels of both calcium and vitamin D. Encourage weight-bearing exercise, minimize modifiable risk factors (e.g., smoking, low BMI, use of steroids). Use TAF instead of TDF or from LPV/r to EFV or an un-boosted INSTI whenever possible

Clinical Adverse Event and Presentation	Signs and Symptoms	Associated ARV	Drug Management
Rash	Most rashes are mild to moderate, diffuse maculopapular eruptions. A rash can be the initial manifestation of systemic hypersensitivity (see the SJS/TEN/EM Major and HSR sections).	Any ARV or TMP-SMX	Mild-to-Moderate Maculopapular Rash Without Systemic or Mucosal Involvement: Most rashes will resolve without intervention; ARV drugs can be Continued while monitoring. Antihistamines may provide some relief. Severe Rash and/or Rash Accompanied by Systemic Symptoms: Manage as SJS/TEN/EM major, DRESS, or HSR as applicable Rash in Patients Receiving NVP: Given the elevated risk of HSR, measure hepatic transaminases. If hepatic transaminases are elevated, NVP should be discontinued and not Restarted (see the HSR section below).

Clinical Adverse Event and Presentation	Signs and Symptoms	Associated ARV	Drug Management
DRESS	Fever; adenopathy; Facial swelling; morbiliform to polymorphous rash; Peripheral eosinophilia; atypical circulating lymphocytes; Internal organ involvement especially the liver and kidneys	DRV, DTG, EFV, ETR, NVP, RAL, RPV	CBC and AST, ALT, and creatinine levels Discontinue all ARV drugs and other possible causative agents (e.g., TMP/SMX). The role of systemic steroids in treatment unclear; consultation with a specialist is recommended. Provide supportive care for the end-organ disease. Do not reintroduce the offending medication.
HSR with or without skin involvement and including SJS/TEN	Symptoms include high fever, diffuse skin rash, malaise, nausea, headache, myalgia, arthralgia, diarrhea, vomiting, abdominal pain, pharyngitis, and respiratory symptoms (e.g., dyspnea). With the continuation of ABC, symptoms	ABC	Counsel patients and families about the signs and symptoms of HSR to ensure prompt reporting of reactions Discontinue ARV drugs and investigate other causes of the symptoms

Clinical Adverse Event and Presentation	Signs and Symptoms	Associated ARV	Drug Management
HSR Cont'd	may progress to hypotension and vascular collapse. With challenge, symptoms can mimic anaphylaxis.	ABC	(e.g., a concurrent viral illness). Provide symptomatic treatment. Most symptoms resolve within 48 hours after discontinuing ABC. Do not re-challenge with ABC
HSR with or without skin involvement and excluding SJS/TEN	Flu-like symptoms (including nausea, vomiting, myalgia, fatigue, fever, abdominal pain, and jaundice) with or without skin rash that may progress to hepatic failure with encephalopathy. NVP hepatotoxicity and HSR are less common in prepubertal children than in adults, and both are uncommon in infants.	NVP	Avoid NVP use in women with CD4 counts >250 cells/mm ³ and in men with CD4 counts >400 cells/mm ³ , unless benefits outweigh risks. Do AST and ALT levels in patients with rash. Discontinue ARV drugs. Consider other causes for hepatitis and discontinue all hepatotoxic medications. Provide supportive care as indicated

Clinical Adverse event and presentation	Signs and Symptoms	Associated ARV	Drug Management
HSR Cont'd		NVP	Avoid the NNRTI class when restarting
SJS/TEN/EM major	The initial rash may be mild, but it often becomes painful, evolving to blister/ bulla formation with necrosis in severe cases. Usually involves mucous membrane ulceration and/ or conjunctivitis. Systemic symptoms may also include fever, tachycardia, malaise, myalgia and arthralgia.	Many ARTS especially NNRTI	Discontinue all ARV drugs and other agents, e.g., TMP/ SMX. Provide intensive supportive, including IV hydration, aggressive wound care, eye care, labial adhesion preventative care, pain management, and antipyretics. Parenteral nutrition and antibiotics may also be necessary. Corticosteroids and/ or Intravenous Immuno-globulins are sometimes used, but the use of these interventions is controversial. Do not reintroduce the offending medication. When SJS/TEN/EM major occurs with the use of one NNRTI avoid other NNRTIs.

Appendix 10: Opportunistic Infections

Treatment of Opportunistic Infections (OI)

Indication	Treatment
Bacterial pneumonia S. pneumoniae; S. aureus, H.influenzae, Aeruginosa	Ceftriaxone 50–100 mg/kg body weight per dose od (max 4–6 g/day IV Cefuroxime, 35–50 mg/kg body weight per dose tds (max 4 g/day) IV For children who are receiving effective ARV, have mild or no immunosuppression, and have mild to moderate community-acquired pneumonia, oral therapy option would be amoxicillin 45 mg/kg body weight per dose twice daily (maximum dose: 4 g per day. Add azithromycin for hospitalized patients to treat other common community-acquired pneumonia pathogens (<i>M. pneumoniae</i>, <i>C. pneumoniae</i>. Add clindamycin or vancomycin if methicillin-resistant <i>S. aureus</i> is suspected (base the choice on local susceptibility patterns. For patients with neutropenia, chronic lung disease other than asthma (e.g., bronchiectasis or indwelling venous catheter, consider a regimen that includes activity against <i>P. aeruginosa</i> (such as ceftazidime or cefepime instead of ceftriaxone. Consider PCP in patients with severe pneumonia or more advanced HIV disease. Evaluate for tuberculosis, cryptococcosis, and endemic fungi as epidemiology suggests these may be possibilities.

Indication	Treatment
Candida	Invasive Disease: • Fluconazole 12 mg/kg body weight IV once daily (maximum 600 mg/day) for a minimum 2 weeks after last positive blood culture (if uncomplicated candidemia) • Lipid formulations of amphotericin B, 5 mg/kg body weight IV once daily • Amphotericin B deoxycholate, 1 mg/kg body weight IV once daily If available Anidulafungin and capsosungin are preferred agents <u>Not critically ill</u> Fluconazole 12 mg/kg body weight/dose daily IV (maximum dose: 600 mg) for infants and children of all ages • Avoid fluconazole for <i>C. krusei</i> and <i>C. glabrata</i>, avoid echinocandin for <i>C. parapsilosis</i>. Treatment Duration: • Based on the presence of deep-tissue foci and clinical response; in patients with candidemia, treat until 2 weeks after last positive blood culture.
Cryptosporidiosis	There is no consistently effective therapy for cryptosporidiosis in HIV infected individuals. Effective ARV may lead to microbiologic and clinical response A trial of nitazoxanide can be considered with optimized ARV • 1–3 years: Nitazoxanide (20mg/mL oral solution) 100 mg orally twice daily with food • 4–11 years: Nitazoxanide (20 mg/mL oral solution) 200 mg orally twice daily with food

Indication	Treatment
	• ≥ 12 years: Nitazoxanide tablet 500 mg orally twice daily with food Treatment Duration: • 3–14 days Supportive Care: • Hydration, correct electrolyte abnormalities, nutritional support • Antimotility agents (such as loperamide) should be used with caution in young children.
Giardiasis Note: Based on data from HIV uninfected children	• Tinidazole, 50 mg/kg by mouth, administered as 1 dose given with food (maximum 2 g). Nitazoxanide. • 1–3 years: 100 mg by mouth every 12 hours with food for 3 days • 4–11 years: 200 mg by mouth every 12 hours with food for 3 days • ≥ 12 years: 500 mg by mouth every 12 hours with food for 3 days Metronidazole 5 mg/kg by mouth every 8 hours for 5–7 days is an alternative choice Supportive Care: • Hydration • Correction of electrolyte abnormalities • Nutritional support • Antimotility agents (e.g., loperamide) should be used with caution in young children.

Indication	Treatment
Herpes Simplex Virus	<u>Neonatal CNS or Disseminated Disease:</u> • Acyclovir 20 mg/kg body weight IV/dose every 8 hours for ≥ 21 days. Acyclovir should not be discontinued until repeat CSH HSV DNA PCR negative <u>Neonatal Skin, Eye, or Mouth Disease:</u> • Acyclovir 20 mg/kg body weight IV/dose every 8 hours for 14 days <u>CNS or Disseminated Disease in Children Outside the Neonatal Period:</u> • Acyclovir 10 mg/kg body weight (up to 15 mg/kg body weight/dose in children <12 years) IV every 8 hours for 21 days <u>Moderate to Severe Symptomatic Gingivostomatitis:</u> • Acyclovir 5–10 mg/kg body weight/dose IV every 8 hours. Patients can be switched to oral therapy after lesions have begun to regress and therapy continued until lesions have resolved. <u>Mild Symptomatic Gingivostomatitis:</u> • Acyclovir 20 mg/kg body weight (maximum 400 mg/dose) dose by mouth QID for 7–10 days Recurrent Herpes Labialis: • Acyclovir 20 mg/kg body weight (maximum 400 mg/dose) dose by mouth QID for 5 days <u>For First-Episode Genital Herpes (Adults and Adolescents):</u> • Acyclovir 20 mg/kg body weight (maximum 400 mg/dose) dose by mouth TID for 7–10 days

Indication	Treatment
	<u>Recurrent Genital Herpes (Adults and Adolescents):</u> • Acyclovir 20 mg/kg body weight (maximum 400 mg/dose) dose by mouth TID for 5 days Children with HSV <u>Keratoconjunctivitis:</u> • Often treated with topical trifluridine (1%) or ganciclovir (0.15%) applied as 1–2 drops 5 times daily. Add oral acyclovir to the topical therapy and refer ophthalmologist <u>Children with Acute Retinal Necrosis</u> • For children old enough to receive an adult dose, acyclovir 10–15 mg/kg body weight/dose IV every 8 hours for 10–14 days, followed by oral valacyclovir 1 g/dose TID for 4–6 weeks OR oral acyclovir 20 mg/kg body weight/dose QID for 4–6 weeks. Refer Ophthalmologist
Isosporiasis (Cystoisosporiasis)	TMP-SMX 5 mg/kg body weight of the TMP component (maximum 160 mg TMP) twice daily by mouth for 10 days If symptoms worsen or persist, the TMP-SMX dose (5 mg/kg/dose of the TMP component) may be given more frequently (e.g., 3–4 times daily by mouth for 10 days) and/or the duration of treatment may be increased to 3–4 weeks. Alternative drugs include Nitazoxanide (see doses above for 3 consecutive days) or Ciprofloxacin 10–20 mg/kg body weight (maximum 500 mg) by mouth twice daily for 7 days (not recommended in children due to risk of adverse events involving the joints and surrounding tissues)

Indication	Treatment
Mycobacterium Avium Complex (MAC)	Initial Treatment (≥ 2 Drugs): • Clarithromycin 7.5–15 mg/kg body weight (maximum 500 mg/ dose) orally twice daily plus ethambutol 15–25 mg/kg body weight (maximum 2.5g/day) orally once daily followed by chronic suppressive therapy. For Severe Disease, add: • Rifabutin 10–20 mg/kg body weight (maximum 300 mg/day) orally once daily OR • Ciprofloxacin 10–15 mg/kg orally twice daily (maximum 1.5 g/day) daily (not labeled for children < 18 years. Consult specialist), OR • Levofloxacin 500 mg orally once daily (not labeled for children < 18 years. Consult specialist) OR • Amikacin 15–30 mg/kg body weight IV in 1 or 2 divided doses (maximum 1.5 g/day) Combination therapy with a minimum of 2 drugs is recommended for ≥ 12 months Chronic suppressive therapy (secondary prophylaxis) is recommended in children and adults following initial therapy. Children receiving ethambutol who are old enough to undergo routine eye testing should have monthly monitoring of visual acuity and colour discrimination.
Mycobacterium Tuberculosis	Intrathoracic Disease Drug-Susceptible TB Intensive Phase (2 Months): • Isoniazid, 10–15 mg/kg body weight (maximum 300 mg/day) by mouth once daily, plus • Rifampin 10–20 mg/kg body weight (maximum 600 mg/day) by mouth once daily, plus

Indication	Treatment
	- Pyrazinamide 30–40 mg/kg body weight (maximum 2g/day) by mouth once daily, plus. - Ethambutol 15–25 mg/kg body weight (maximum 2.5 g/day) by mouth once daily. Continuation Phase (4 Months): - Isoniazid 10–15 mg/kg body weight (maximum 300 mg/day) by mouth once daily, plus - Rifampin 10–20 mg/kg body weight (maximum 600 mg/day) by mouth once daily Therapy should be extended from 4 to 7 months for a total of 9 months of therapy if there is: - A positive sputum culture after 2 months of treatment - A cavity on chest x-ray at the end of treatment - More extensive disease and/or cavities on a chest x-ray in the first 2 months of therapy As children approach a body weight of 25 kg, clinicians can use adult dosing recommendations. Extra thoracic Disease: - Lymph node with TB—treat as minimal intrathoracic disease - Bone or joint disease— extend the continuation phase to 10 months (for total duration of therapy of 12 months). TB Meningitis - Extend the continuation phase to 10 months (for total duration of therapy of 12 months). - Discuss with an expert. Only DOT should be administered.

Indication	Treatment
	Adjunctive Treatment: • Co-trimoxazole prophylaxis • Pyridoxine 1–2 mg/kg/ body weight/day (maximum 25–50mg/day) with isoniazid • Corticosteroids (2 mg/kg body weight per day of prednisone [maximum, 60 mg/day] for 4–6 weeks followed by tapering) with CNS disease or pericardial effusion; may be considered with pleural effusions, severe airway compression, or severe IRIS. • Nutrition Support If ARV-naive, start TB therapy immediately and initiate ARV within 2–8 weeks. (See ARV treatment chapter)
Pneumocystis Pneumonia	TMP-SMX 3.75–5 mg/kg body weight/dose TMP (based on TMP component) every 6 hours IV or orally given for 21 days (followed by secondary prophylaxis dosing) Alternative regime if intolerant or treatment failure is Pentamidine 4 mg/kg body weight/dose IV/IM once daily. After acute pneumonitis resolved in mild / moderate disease, IV TMP-SMX can be changed to oral. For oral administration, total daily dose of TMP-SMX can also be administered in 3 divided doses (every 8 hours). Indications for Corticosteroids: • $\text{PaO}_2 < 70$ mm Hg at room air or alveolar arterial oxygen gradient > 35 mmHg Prednisone Dose: • 1 mg/kg body weight/dose by mouth twice daily for 5 days, then • 0.5–1 mg/kg body weight/dose by mouth twice daily for 5 days, then

Indication	Treatment
	0.5 mg/kg body weight by mouth once daily for days 11 to 21. Chronic suppressive therapy (secondary prophylaxis) with TMP/SMX is recommended in children and adults following initial therapy.
Syphilis	<i>Congenital Proven or Highly Probable Disease:</i> - Aqueous crystalline penicillin G 100,000–150,000 units/kg body weight per day, administered as 50,000 units/kg body weight per dose IV every 12 hours for the first 7 days of life, and then every 8 hours for 10 days - If diagnosed after 1 month of age, aqueous penicillin G 200,000–300,000 unit/kg body weight per day, administered as 50,000 units/kg body weight per dose IV every 4–6 hours (maximum 18–24 million units per day) for 10 days <i>Possible Disease- Acquired:</i> Early Stage (Primary, Secondary, Early Latent): - Benzathine penicillin 50,000 units/kg body weight (maximum 2.4 million units) IM for 1 dose <i>Late Latent:</i> - Benzathine penicillin 50,000 units/kg body weight (maximum 2.4 million units) IM once weekly for 3 doses <i>Neurosyphilis (Including Ocular):</i> - Aqueous penicillin G 200,000–300,000 units/kg body weight per day administered as 50,000 units/kg body weight per dose IV every 4–6 hours (maximum 18–24 million units per day) for 10–14 days

Indication	Treatment
	Examinations and serologic testing for children with congenital syphilis should occur every 2–3 months until the test becomes non-reactive or there is a fourfold decrease in titer. Children with increasing titers or persistently positive titers (even if low levels) at ages 6–12 months should be evaluated and considered for re-treatment. Children and adolescents with acquired syphilis should have clinical and serologic response monitored at 3, 6, 9, 12, and 24 months after therapy.
Toxoplasmosis	<i>Congenital Toxoplasmosis:</i> • Pyrimethamine loading dose—2 mg/kg body weight by mouth once daily for 2 days, then 1 mg/kg body weight by mouth once daily for 2–6 months, then 1 mg/kg body weight by mouth 3 times weekly, plus • Leucovorin (folinic acid) 10 mg by mouth or IM with each dose of pyrimethamine, plus • Sulfadiazine 50 mg/kg body weight by mouth twice daily <i>Treatment Duration: 12 months</i> Acquired Toxoplasmosis <i>Acute Induction Therapy (Followed by Chronic Suppressive Therapy):</i> • Pyrimethamine: loading dose—2mg/kg body weight (maximum 50mg) by mouth once daily for 3 days, then 1mg/kg body weight (maximum 25 mg) by mouth once daily, plus • Sulfadiazine 25–50 mg/kg body weight (maximum 1–1.5 g/dose) by mouth per dose 4 times daily, plus • Leucovorin 10–25 mg by mouth once daily, followed by chronic suppressive therapy

Indication	Treatment
	<i>Treatment Duration (Followed by Chronic Suppressive Therapy):</i> • ≥6 weeks (longer duration if the clinical or radiologic disease is extensive or response is incomplete in 6 weeks) • Pyrimethamine use requires CBC monitoring at least weekly while on daily dosing and at least monthly while on less than daily dosing. Corticosteroids (e.g., prednisone, dexamethasone) have been used in children with CNS disease when CSF protein is very elevated (>1,000mg/dL), or there are focal lesions with significant mass effects, with discontinuation as soon as clinically feasible. • Anticonvulsants should be administered to patients with a history of seizures and continued through the acute treatment, but should not be used prophylactically.
Varicella-Zoster	Chickenpox <i>Children with No or Moderate Immune Suppression (CDC Immunologic Categories 1 and 2) and Mild Varicella Disease:</i> • Acyclovir 20 mg/kg body weight/dose by mouth (max 800 mg/dose) QID for 7–10 days and until no new lesions for 48 hours <i>Children with Severe Immune Suppression (CDC Immunologic Category 3):</i> • Acyclovir 10 mg/kg body weight 500 mg/m²/dose IV every 8 hours for 7–10 days and until no new lesions for 48 hours <i>Zoster Children with Uncomplicated Zoster:</i> • Acyclovir 20 mg/kg body weight/dose (max 800 mg/dose) by mouth QID for 7–10 days.

Indication	Treatment
	Children with Severe Immunosuppression (<i>CDC Immunologic Category 3</i>), Trigeminal or Sacral Nerve Involvement, Extensive Multidermatomal, or Disseminated Zoster: - Acyclovir 10 mg/kg body weight/dose IV every 8 hours until cutaneous lesions and visceral disease are clearly resolving, then can switch to acyclovir by mouth to complete a 10- to 14-day course <i>Children with Progressive Outer Retinal Necrosis:</i> - Ganciclovir 5 mg/kg body weight/dose IV every 12 hours, plus - Foscarnet 90 mg/kg body weight/dose IV) every 12 hours, plus - Ganciclovir 2 mg/0.05 mL intravitreal twice weekly and/or Foscarnet 1.2 mg/0.05 mL intravitreal twice weekly <i>Children with Acute Retinal Necrosis:</i> - Acyclovir 10–15 mg/kg body weight/dose IV every 8 hours daily for 10–14 days, followed by oral acyclovir dose: 20 mg/kg body weight/dose QID for 4–6 weeks OR Oral valacyclovir 1 g/dose TID for 4–6 weeks (for children old enough to receive adult dose).

Guidelines for the Prevention and Treatment of Opportunistic Infections in HIV-Exposed and HIV-Infected Children. https://aidsinfo.nih.gov/contentfiles/lvguidelines/oi_guidelines_pediatrics.pdf

Appendix 11: Verbal and Non-verbal Listening and Learning Skills

The following are 7 verbal and non-verbal listening and learning skills to help the client explore, understand and take action:

- Skill 1: Use helpful non-verbal communication
- Skill 2: Actively listen and show interest in the client
- Skill 3: Ask open-ended questions
- Skill 4: Paraphrasing what the client is saying
- Skill 5: Empathize — show that you understand how the client feels
- Skill 6: Avoid words that sound judging
- Skill 7: Help the client set goals and summarize each counselling session

Skill 1: Using helpful non-verbal communication

Non-verbal communication includes messages conveyed through gestures, gaze, posture, and facial expressions. This encourages the client to feel that the health worker is listening and cares about what is being said. The acronym “ROLES” can be used to help remind health workers of behaviours that convey caring

Non-verbal behaviours that convey caring	
R	A relaxed and natural attitude with clients is important. Do not move around quickly or chat nervously.
O	Adopt an open posture . Crossing your legs or arms can signal that you are critical of what the client is saying or that you are not listening. Using an open posture shows that you are open to the client and what he or she is saying.
L	Leaning forward toward the client is a natural sign of involvement.
E	Culturally appropriate eye contact should be maintained to communicate interest. Never stare or glare at the client.
S	Sitting squarely facing another person shows involvement. If, for any reason, this may be considered threatening, then sitting to the side is also an option.

Skill 2: Actively listen and show interest in the client

Active listening shows that you are interested and want to encourage a client to talk by using gestures like nodding and smiling, responses like “Mmm” or “Aha”.

Clarifying prevents misunderstanding and helps sort out what has been said. For example, if an adolescent says: “*All my friends will abandon me if they find out I have HIV!*” the health worker could say, “*Tell me more about why disclosing to your friends is a concern for you.*”

Summarizing means pulling together the themes that have come up during a counselling discussion so the client can see the whole picture. Summarizing helps ensure that the client and the health worker understand each other.

Example of active Listening

HW: <i>Hey, you seem sad today. What’s going on?</i>
Male adolescent client: <i>It’s school, I don’t want to go anymore.</i>
HW: <i>umhummm*</i> (nods understandingly)
Adolescent: <i>Well, it’s not really school, it’s the other pupils at school...I don’t have any friends.</i>
HW: <i>So, you don’t like going to school because you feel like you don’t fit in? **</i>
Adolescent: <i>Yeah, the other children make fun of me. They call me mean names.</i>
HW: <i>umhummm*</i>
Adolescent: <i>Yesterday, one of the bigger boys even pushed me to the ground and tried to take my pocket money.</i>
HW: <i>That’s terrible. It seems to me that the other boys are harassing you. What one thing would you like to change to make this situation better? ***</i>
* A gesture that shows interest ** Clarifying *** Summarizing

Skill 3: Ask open-ended questions

Open-ended questions begin with words like *“how,” “what,” “when,” “where,”* or *“why.”* This type of question encourages the client to talk openly and in a way that leads to further discussion. They help clients explain their feelings and concerns, and they also help counselors get the information they need to help clients make decisions.

Avoid using questions that require a yes or no answer. They usually start with words like, *“are you?” “did he?” “has she?”* or *“do you?”*.

Closed-Ended	Open-Ended
<i>“Are you concerned about talking to your family about your diagnosis?”</i>	<i>“What concerns do you have about talking to your family about your HIV test results?”</i>
<i>“Are you taking your ARVs?”</i>	<i>“How many times have you taken your ARVs in the last 3 days?”</i>
	<i>“Which doses did you miss?”</i>
	<i>“What problems have you had taking your ARVs lately?”</i>

Very sensitive questions are easier to ask as close-ended rather than open-ended questions.

For example, when asking if a client is sexually active, it may be offensive to the inexperienced client if you ask, *“When was the last time you had sex?”*

Instead, it may be more appropriate to say, *“Many adolescents have sex with their partners. Are you having sex?”* even though this is a closed-ended question.

Skill 4: Paraphrasing what the client is saying

This involves repeating what the client says in a slightly different way to encourage them to say more.

For example, if a client says, *“I can’t tell my boyfriend about my HIV-status,”* the health worker could reflect by saying, *“It sounds like talking to your partner is not something that you feel comfortable doing right now.”* After the client confirms that this is accurate, the health worker could then say, *“Let’s talk about that some more.”*

Health workers can use the following formulas for reflecting:

- *“You feel _____ because _____.”*
- *“You seem to feel that _____ because _____.”*
- *“You think that _____ because _____.”*
- *“So I sense that you feel _____ because _____.”*
- *“I’m hearing that when _____ happened, you didn’t know what to do.”*

Skill 5: Empathize — show that you understand how the client feels

Showing empathy helps encourage clients to discuss issues further.

Client says	Empathetic response
1. <i>“I just can’t tell my partner that I have HIV!”</i>	<i>“It sounds like you might be afraid of your partner’s reaction.”</i>
2. <i>“My partner argues with me all the time about using condoms! I’m so sick of fighting with him,”</i>	<i>“That must be really upsetting. It sounds like you feel very frustrated with him.”</i>

If the health worker responds to number 2 with a factual question like, *“How often do you have these kinds of fights about condoms?”* the client may feel that the health worker does not understand because his or her response was not in reference to the client’s underlying feelings.

When empathizing, the health worker identifies and articulates the emotions behind a client’s statement. This is different from “Skill 4: Paraphrasing what the client is saying,” which is used to summarize a conversation that is primarily factual.

Skill 6: Avoid words that sound judging

Examples of judging words	Alternative open-ended non judging words
HW: <i>Did you listen to me and use a condom?</i> Client: <i>Um...yes.</i>	<i>What form of family planning, if any, did you use the last time you had sex?</i>
HW: <i>Did you take your medicine correctly (or properly)?</i> Client: <i>I think so.</i>	<i>How many times in the last 3 days have you taken your ARVs?</i>
HW: <i>Didn't you understand what I told you about taking your medicine?</i> Client: <i>I don't know, I think so.</i>	<i>When do you usually take your medicines? How do you take them (with water? with food? etc.)?</i>
HW: <i>Did you do the right thing and talk to your mother your HIV-Status?</i> Client: <i>Well, yes, I tried to talk to her....</i>	<i>What has been your experience with taking ARV? What problems, if any, have you had taking your ARVs?</i> <i>Can we go back to our discussion about disclosure? Who have you told about your HIV test result since your last visit?</i>

Judging words, when used in asking questions, may cause the adolescent client to feel that they are wrong and give a misleading response for fear of being judged. Additionally, an opportunity to understand and address the situation in more detail may be lost as the client becomes uncomfortable and unwilling to share valuable information.

Skill 7: Help the client set goals and summarize each counselling session

• Work with the client to come up with realistic “next steps.” To help the client develop a more specific plan, the health worker could ask: • <i>What do you think might be the best thing to do?</i> • <i>What will you do now?</i> • <i>How will you do this?</i> • <i>Who might help you?</i> • <i>When will you do this?</i>
• Summarize the main points of the counselling session. • The health worker could say, <i>“I think we’ve talked about a lot of important things today. (List main points.) We agreed that the best next steps are to _____.</i> <i>Does that sound right? Let’s plan a time to talk again soon.”</i>
• Give the client a chance to ask questions. • Make referrals, if needed. • Set a next appointment date and discuss the availability of clinic services outside of clinic visits.

Common Counselling Scenarios

Scenario	What the health worker can do
• Silence	• Remember that silence can be a sign of shyness, embarrassment, anger, or anxiety. • If an adolescent client is silent at the beginning of a session, the health worker can say, “I realize it’s hard for you to talk. Talking to someone you don’t know can be scary. Many people are scared (or too embarrassed, too angry, too anxious) to share their feelings.”

Scenario	What the health worker can do
Anger	Say, <i>"You seem angry. It's OK to be angry, but would you like to talk about it?"</i> Or, if the health worker thinks he or she knows why the client is angry, he or she can say something like: "Sometimes when someone comes to see me against his or her will and doesn't want to be here, it is difficult for him, or her to speak. Is that what is going on?"
Shyness	- Legitimize the feeling by saying, <i>"I would feel the same way in your place. I understand that it's not easy to talk to a person you have just met."</i> - Use books, brochures, or posters to encourage discussion or refer to a story or anecdote so the adolescent can talk about others rather than him- or herself. Some adolescents need time to feel comfortable with someone new. - If the adolescent cannot or will not talk, the health worker should propose another meeting.
Crying	- Try to evaluate what provoked the tears and assess if it makes sense in the given situation. - If the client is crying to relieve tension, the health worker can give the adolescent permission to express his or her feelings by saying, <i>"It's okay to cry...it's the normal thing to do when you're sad."</i> - If the client is using crying as manipulation, the health worker can say, <i>"Although I'm sorry you feel sad, it's good to express your feelings."</i> - The health worker should allow the client to freely express emotions and should not try to stop his or her feelings or belittle their importance.

Scenario	What the health worker can do
• Threat of suicide	• Take all suicide threats very seriously! Refer the adolescent to a qualified counselor, psychiatrist, or psychologist and accompany him or her to the appointment. Work together with relevant members of the multidisciplinary care team to form an appropriate plan of action.
• Refusal of help	• Discreetly try to find out why the adolescent is refusing help. If the underlying feeling is anger, refer to some of the suggestions under “Anger” listed above) • If the client has been sent against his or her will, the health worker can say, <i>“I understand how you feel. I’m not sure I can help you, but maybe we could talk for a minute and see what happens.”</i>

Source ICAP

Session particularly with sexually active adolescents

No-Risk	• Hugging • Kissing (even “French kissing,” or kissing with the tongue, carries no risk of HIV transmission) • Holding hands • Rubbing against one other with clothes on • Sharing fantasies • Self-masturbation
Low Risk	• Masturbating your partner or masturbating together, as long as males do not ejaculate near any opening or broken skin of their partner • Using a male or female latex condom during every act of sexual intercourse (penis in vagina, penis in the anus, penis in mouth, etc.) • Using a barrier method for oral sex on a male or female, or for any mouth-to-genitals or mouth-to-anus contact • Sharing sexual toys (rubber penis, vibrators) without cleaning them
Medium Risk	• Oral sex without a latex barrier (some STIs, like gonorrhea, are easily passed through oral sex, while others, like chlamydia, are not. • The risk of HIV transmission through oral sex is generally low, but there is some risk, especially if the person has an STI or cuts/sores in the mouth or on the genitals)
High Risk	• Unprotected (no male or female condom) anal or vaginal sex

Appendix 12 Sexual & Reproductive Health

Questions for the client:

1. Assess knowledge

- *How is HIV transmitted from one person to another?*
- *How can a person prevent the transmission of HIV during sex?*
- *What is your plan to protect your partner from getting HIV when you have sex?*
- *Did you know that even if both partners have HIV, it is important to practice safer sex and use condoms? Do you know why?*

2. Discuss options for sexual risk reduction

There are several ways to reduce the risk of HIV, other STIs, and unwanted pregnancy, including:

- *Abstinence*
- *Intimate touching without an exchange of bodily fluids*
- *Reducing your number of sexual partners*
- *Disclosing your HIV-status and negotiating sexual practices*
- *Correctly and consistently using condoms (for male-female couples, ideally with another form of hormonal contraception)*
- *STI screening and treatment (HIV is transmitted more easily in the presence of other STIs)*
- *Maintaining maximal suppression of HIV through excellent adherence to ARV, if eligible*
- *Avoiding alcohol, marijuana, party drugs, and other substances that impair good judgment and prevention*

3. If an option, discuss abstinence

- *Abstinence means not having sex. If you are abstinent, you cannot get STIs or get re-infected with HIV, and you cannot have an unplanned pregnancy.*
- *Is abstinence an option for you?*
- *If you choose abstinence, you should have a backup plan as well, just in case you change your mind. What will be your backup plan?*

Questions for the client:**4. Discuss condoms**

Demonstrate steps for putting on a condom (male and female) and offer to supply the client with condoms

Help the client improve condom negotiation skills by:

- Responding to the client's questions and concerns
- Reassuring the client that it can be difficult to bring up the topic of condoms with a partner
- Suggesting that he or she discuss condoms BEFORE they are needed (rather than in the heat of the moment)

5. Roleplay to encourage condom use

If a partner asks: *"But you have never suggested, we use condoms before."*

Client can say: *"I went to the clinic today, and my health worker told me that I really need to use condoms for my health and so that we can prevent unintended pregnancy."* (Or, the client may have another reason to explain changing his or her mind.)

If a partner asks: *"You don't love me enough to have sex without a condom?"*

The client can say: *"It is because I love you and I love myself that I want to keep us both safe."*

If a partner says: *"You must want to use a condom because you have been messing around with other people."*

The client can say: *"Before we met, we both had other partners and I want to be sure that neither of us brings anything into this relationship."*

6. Encourage disclosure

Encourage disclosure to partners, work with clients to facilitate the disclosure process, and offer the possibility of meeting with the client and partner together to help the client disclose.

Multi-component interventions for Sexual and Reproductive health and positive living

	Individual	Group
School-based	Encourage ALHIV to go to and stay in school • Adolescents who stay in school will have more opportunities in the future. • In addition to building academic skills, the school also provides adolescents with a chance to make friends and develop life skills.	• Provide essential health-based information through a curriculum-based approach. This will provide young people with the knowledge, attitudes, and skills to make informed decisions about their sexuality and sexual and reproductive health • Increase awareness of HIV treatment interventions and the key role of a support network and adherence to viral suppression • Use adult and peer-led interventions • Link school-based health service to sexual and reproductive health service

	Individual	Group
Community-based	• Link ALHIV with formal or informal life skills training, such as those offered by youth groups, schools, and faith- or community-based organizations • Encourage the ALHIV to stay socially engaged: making new friends and accessing peer support through youth clubs, sports teams, and/or after-school clubs; joining an ALHIV association; talking about things openly	• Generate community support for school and health-based sexual and reproductive interventions through fairs, meetings, church groups, and social marketing • Increase awareness of HIV treatment interventions and the key role of a support network and adherence to viral suppression • Create positive youth development programmes to focus on school retention, academic success; social support; life skill development and peer mentoring • Incorporate adolescent involvement and leadership and peer education strategies • Establish youth-friendly venues in which young people access information and services that address sexual and reproductive health needs

	Individual	Group
Family-Based	Encourage the ALHIV to talk about things openly with family members they can trust	Promote interventions to improve parent-child communication about sexual health and sexuality
Online	Refer to a trusted source of information such as Avert.org: An online resource with easy to understand information on HIV transmission prevention, treatment, care, epidemiology, pathology, politics, and more. Available at: http://www.avert.org HIVFactSheet Download with Google Play	Develop internet-based culturally relevant safe sex, HIV and STI education and behavioural skills-based interventions
Universal Health coverage	Practice healthy habits (see the chapter on Nutrition and healthy living)	- Youth-friendly health services are to be offered by trained health professionals who provide non-judgemental, confidential services - Services should include: HIV testing counselling, treatment care, and support; education, counselling, and provision of contraceptive devices; treatment and prevention of STI; care for sexual and gender-based violence; mental health screening and interventions, and pre and postnatal care and delivery

	Individual	Group
Faith-Based	Talk with ALHIV about their spiritual, religious, and cultural beliefs and practices • Health workers should encourage both ALHIV and their families to continue their regular spiritual, religious, and cultural practices. • This can be a further source of support and counselling to help them face issues related to mental health, disclosure, and positive living.	• Engage churches in health education, preventative strategies while engendering faith-based moral standards

Adapted from Our Future: Lancet Commission on Adolescent Health and well being . www.thelancet.com Vol 387 June 11, 2016, and Adolescent HIV Care and Treatment: A training curriculum for health workers, Colton T at al. Trainer manual, Module 6. ICAP, New York, 2012

Appendix 13 - Social Support for Vulnerable Children

A network of social support agencies is accessible islandwide. Identification of vulnerable families warrants immediate referral to social services

Case conferences should be arranged with all members of the team to discuss families with challenges affecting adherence

Agency	Contact Information
Child Protection Agencies	1. Child Protection and Family Service Agency (CPFSA). [Formerly Child Development Agency (CDA)] Corporate Office: (876) 948-6678/948-2841-2 Regional Office: (876) 948-1145/948-0243 Western Region Regional office: (876) 979-3446/979-1024 North East Region Regional office: Tel: (876) 972-2686 Southern Region Regional office: Tel: (876) 962-2558 2. Office of the Children's Advocate (OCA) 876-967-3225 3. Office of the Children's Registry (OCR)-1888-PROTECT (7768328)

Agency	Contact Information
Counselling Services	• Women's Centre of Jamaica Foundation • Family Life Ministries • Peer Counselling Association of Jamaica • Ministry of Justice Victim Support Unit • National Council on Drug Abuse • Rise Life Management • Ministry of Education, Guidance, and Counselling • National Family Planning Board • National HIV/STI Programme • The Women's Resource and Outreach Centre Ltd
Support Networks	HEART Trust/NTA located island-wide National Youth Service (NYS) 876-978-5347, 876-978-7881 PATH Programme

Appendix 14: Treatment Sites

Parish	Sites	Address & Contact
Kingston	Bustamante Hospital for Children	5 arthur Wint Drive, Kingston 876-926-5721
	Comprehensive Health Centre	55 Slipen Road, Kingston 876-922-3042
	University Hospital of The West Indies	Mona, Kingston 7 876-927-1621
St. Catherine	Spanish Town Hospital	Burke Road, Spanish Town 876-984-3031
St. Thomas	Princess Margaret Hospital	54 Lyssons Road, Morant Bay 876-982-2304
St. Mary	Port Maria Hospital	Trinity, Port Maria 876-994-2277
	Annotto Bay Hospital	Annotto Bay 876-996-2742
St. Ann	St. Ann's Bay Hospital	Seville Road, St. Ann 876-972-2272
St. James	Cornwall Regional Hospital	Mount Salem, Montego Bay 876-952-5100
Westmoreland	Savanna-la-mar Hospital	Barracks Road, Savanna-la-mar 876-955-2533
Clarendon	May Pen Hospital	Manchester Avenue, May Pen 876-986-2528
Manchester	Mandeville General Hospital	32 Hargreaves Avenue, Mandeville 876-962-2067
St. Elizabeth	Black River Hospital	45 High Street, Black River 876-965-2212

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